

CS-663

Digital Image Processing (DIP)

Basic Operations

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Digital Image

- Image representation: an image is a multi-dimensional array
- Array-element values can be represented as integers or floating-point
 - e.g., int8, uint8, int16, int32, float/single, double
- Image file format = **matrix** + **header**
- **Header** contains information like:
 - File format “signature” (specific sequence of first few bytes in file)
 - Array dimensions and size
 - Pixel size units in physical dimensions,
e.g., 1 x 1 x 1.5 mm (MRI image) or km (google maps)
 - Binary-data indicator (yes/no)
 - Binary-data-format indicator (big/little endian)

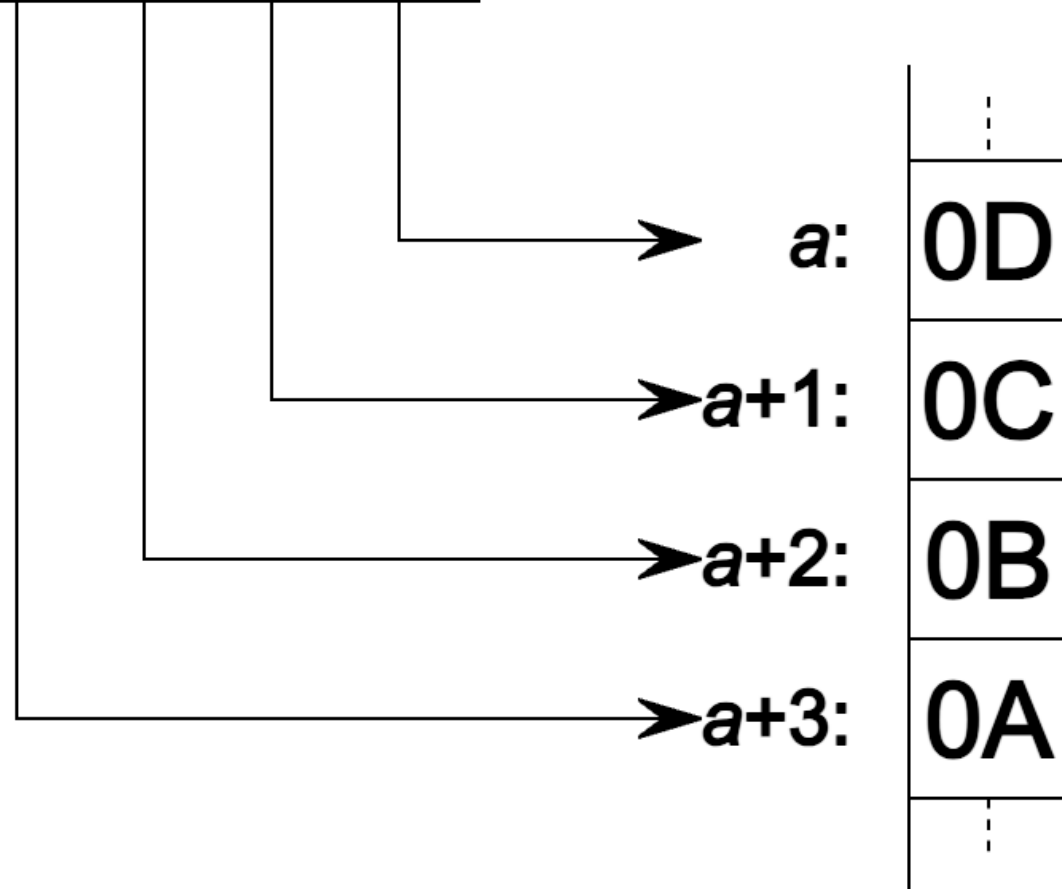
Digital Image

- Endianness: Big-endian format stores most-significant-byte (MSB) at lowest memory address

one 32-bit integer

arranged as
four bytes in
memory

0A0B0C0D

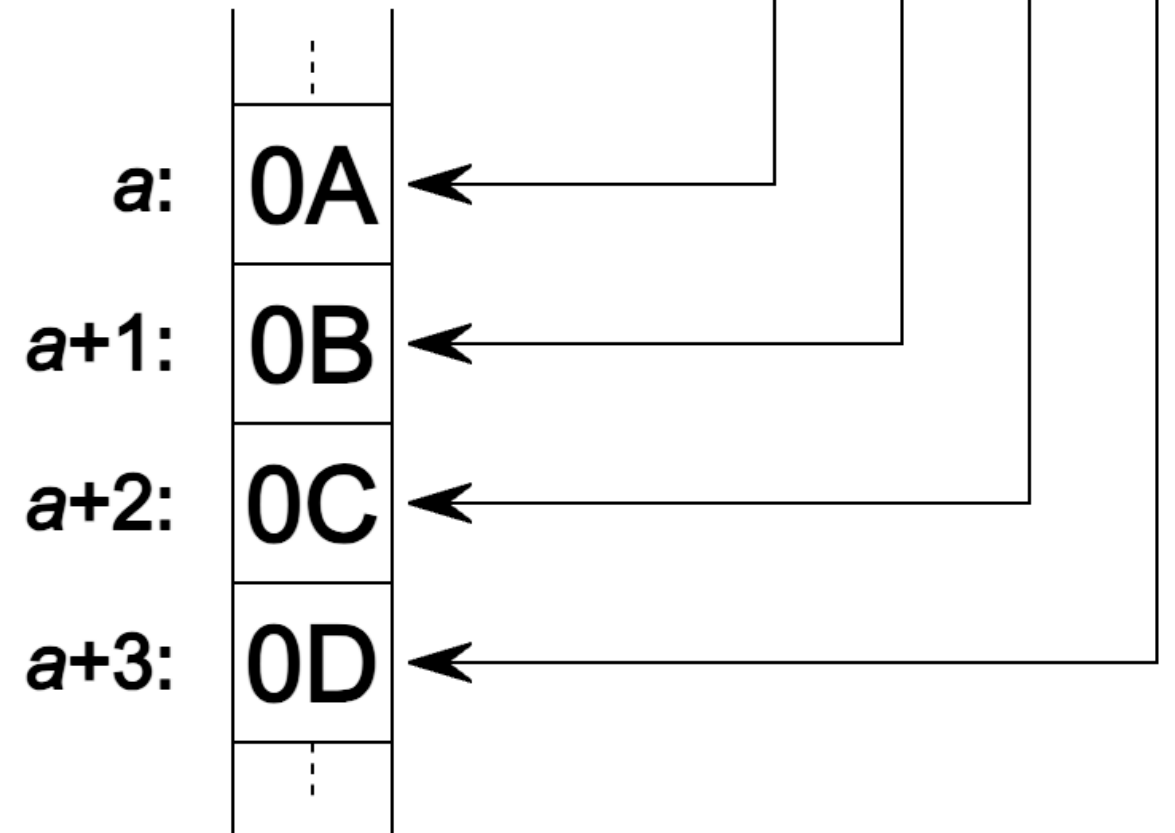


Little-endian

arranged as
four bytes in
memory

one 32-bit integer

0A0B0C0D



Big-endian

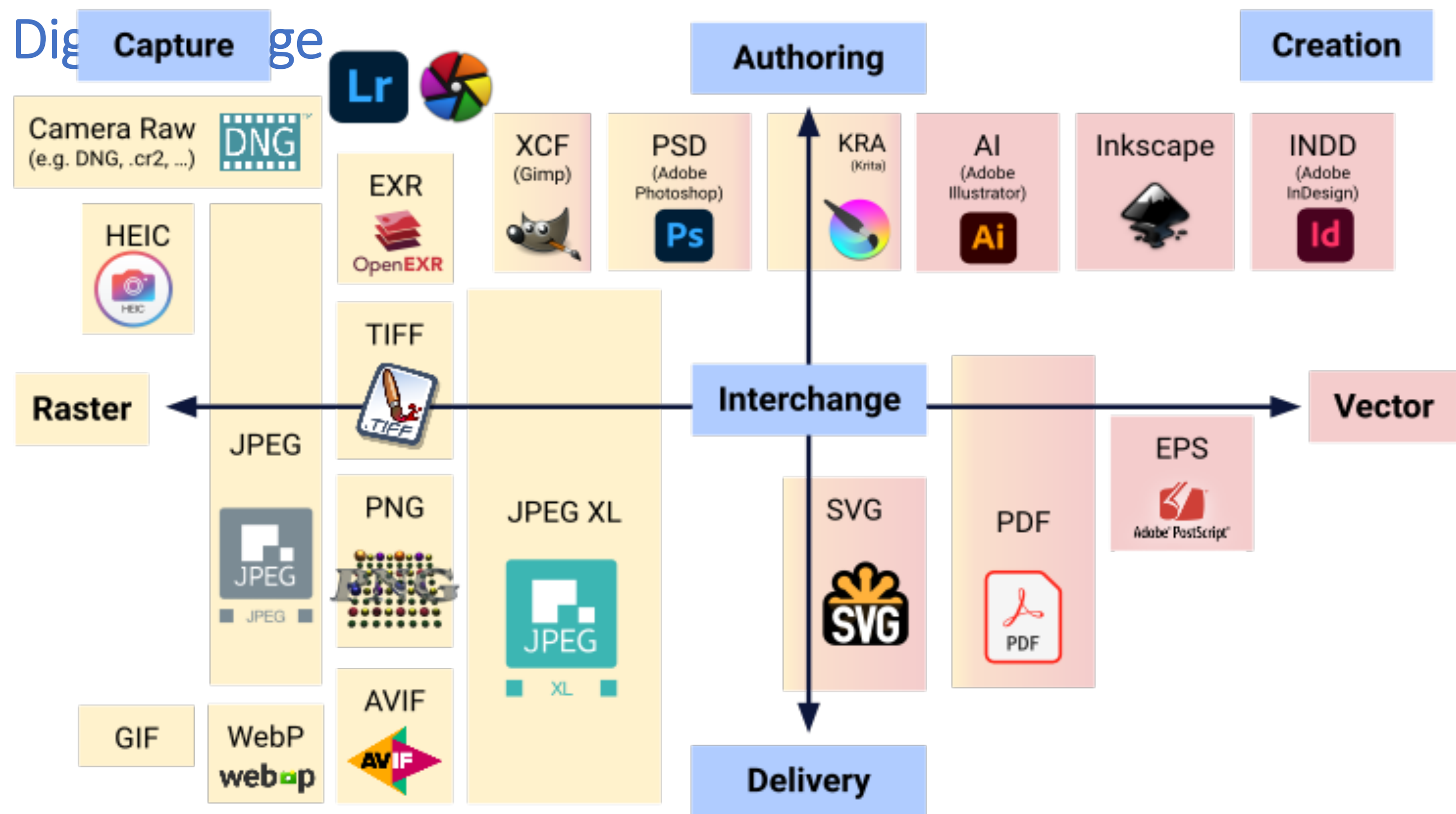
Digital Image

- P2 =
“magic number”
 - Identifies file type
 - Pgm format stores numbers in ASCII
- # → comment
- 17 columns
- 7 rows
- 255 =
maximum value

```
P2
# oie.pgm
17 7
255
0 13 13 13 13 13 13 13 0 0 0 0 0 0 0 0 0
0 13 0 0 0 0 0 13 0 7 7 0 0 81 81 81 81
0 13 0 7 7 7 0 13 0 7 7 0 0 81 0 0 0
0 13 0 7 0 7 0 13 0 7 7 0 0 81 81 81 0
0 13 0 7 7 7 0 13 0 7 7 0 0 81 0 0 0
0 13 0 0 0 0 0 13 0 7 7 0 0 81 81 81 81
0 13 13 13 13 13 13 13 0 0 0 0 0 0 0 0 0
```

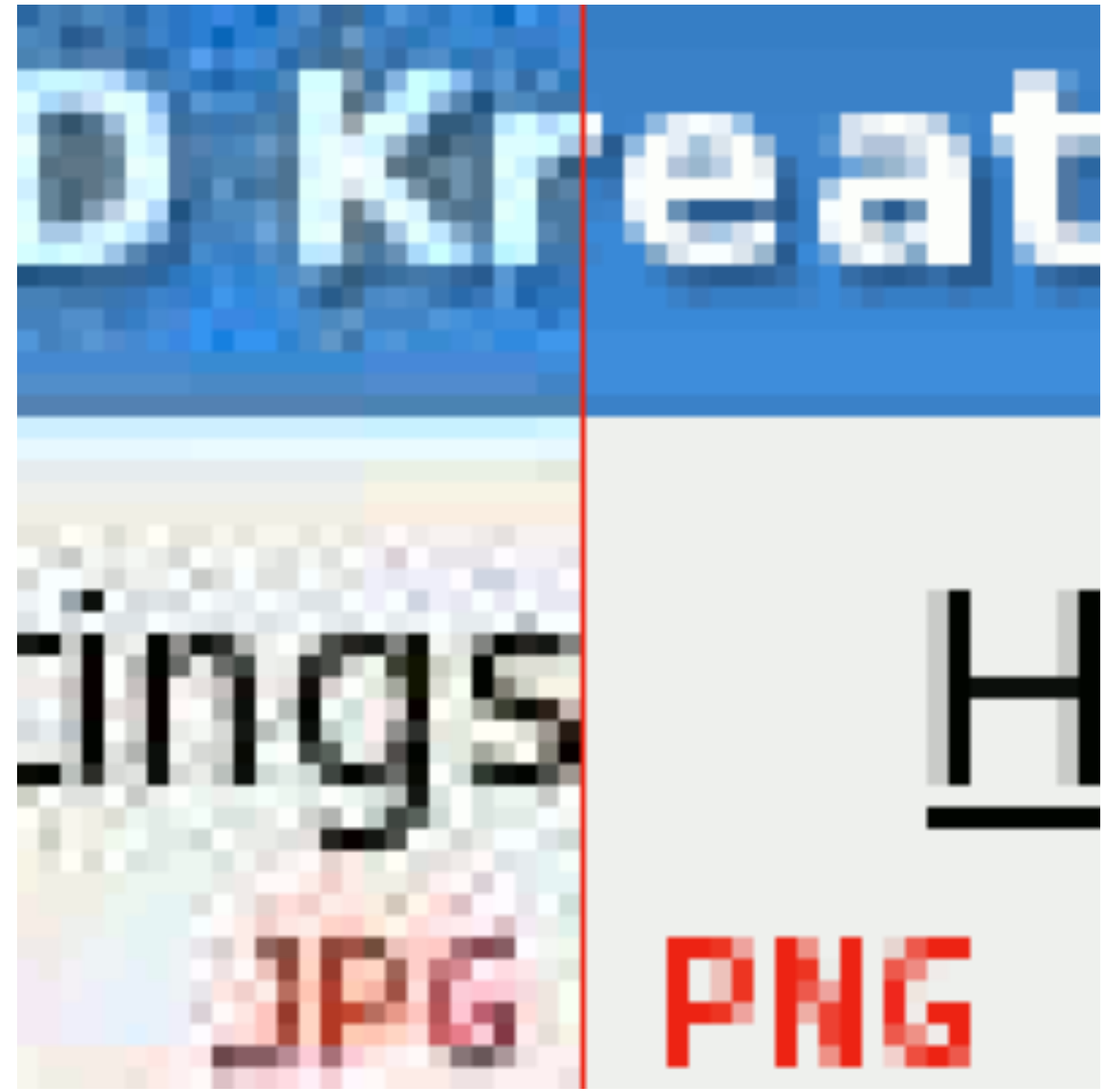


Digital Imaging



Digital Image

- Some image file formats use compression that can lose information
 - e.g., jpeg/jpg, jp2, mpeg, mp4, avi
- **Lossless**/NO compression is great for:
 - Photography: sports, fine art, nature
 - Scientific imaging
 - **All your assignments and course projects (unless otherwise stated)**
 - Format examples: **png, tiff**
OR **npz, mat**



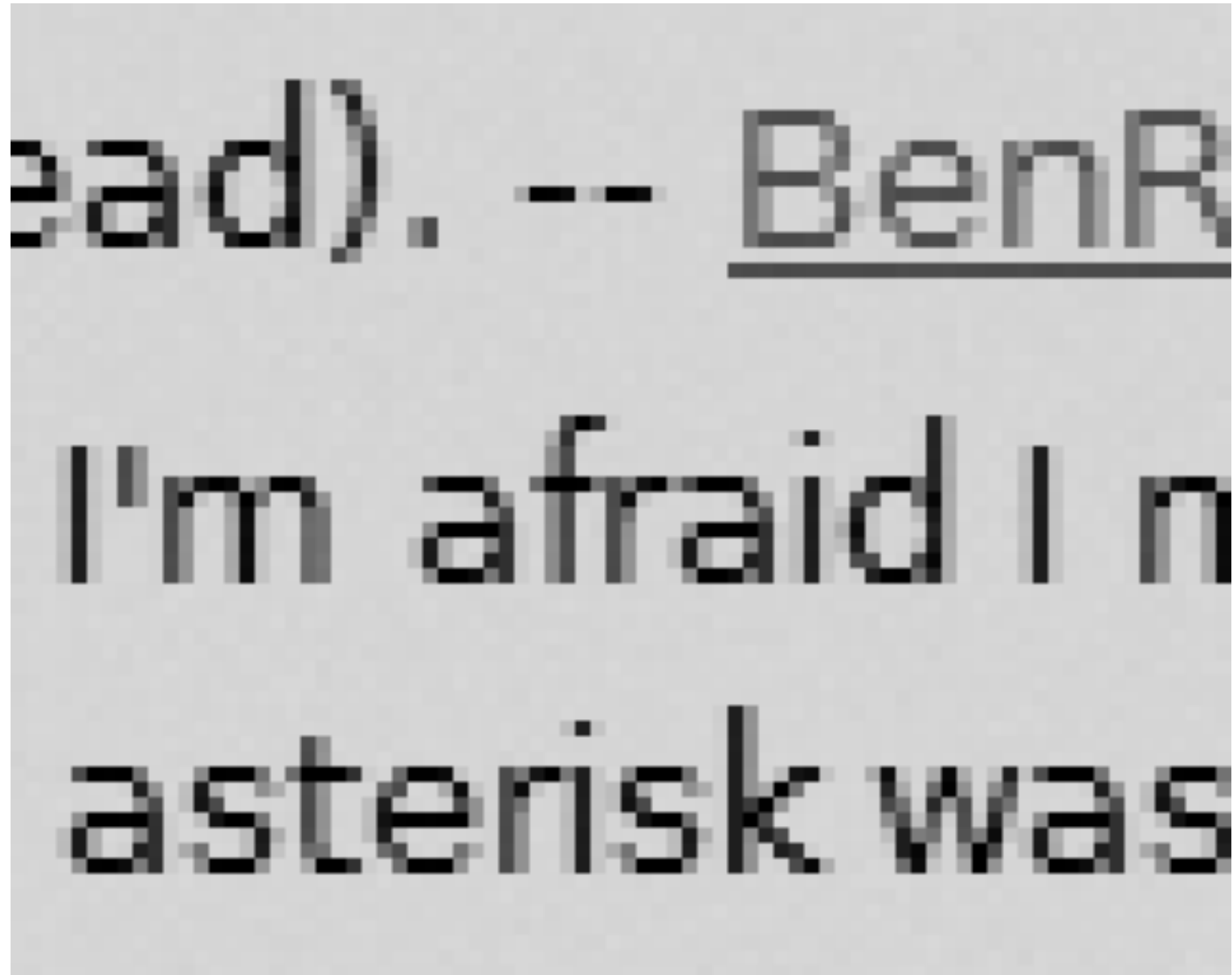
Digital Image

- JPEG Quality:
10, 1,
100, 50, 25
- Even quality 100 isn't
guaranteed lossless



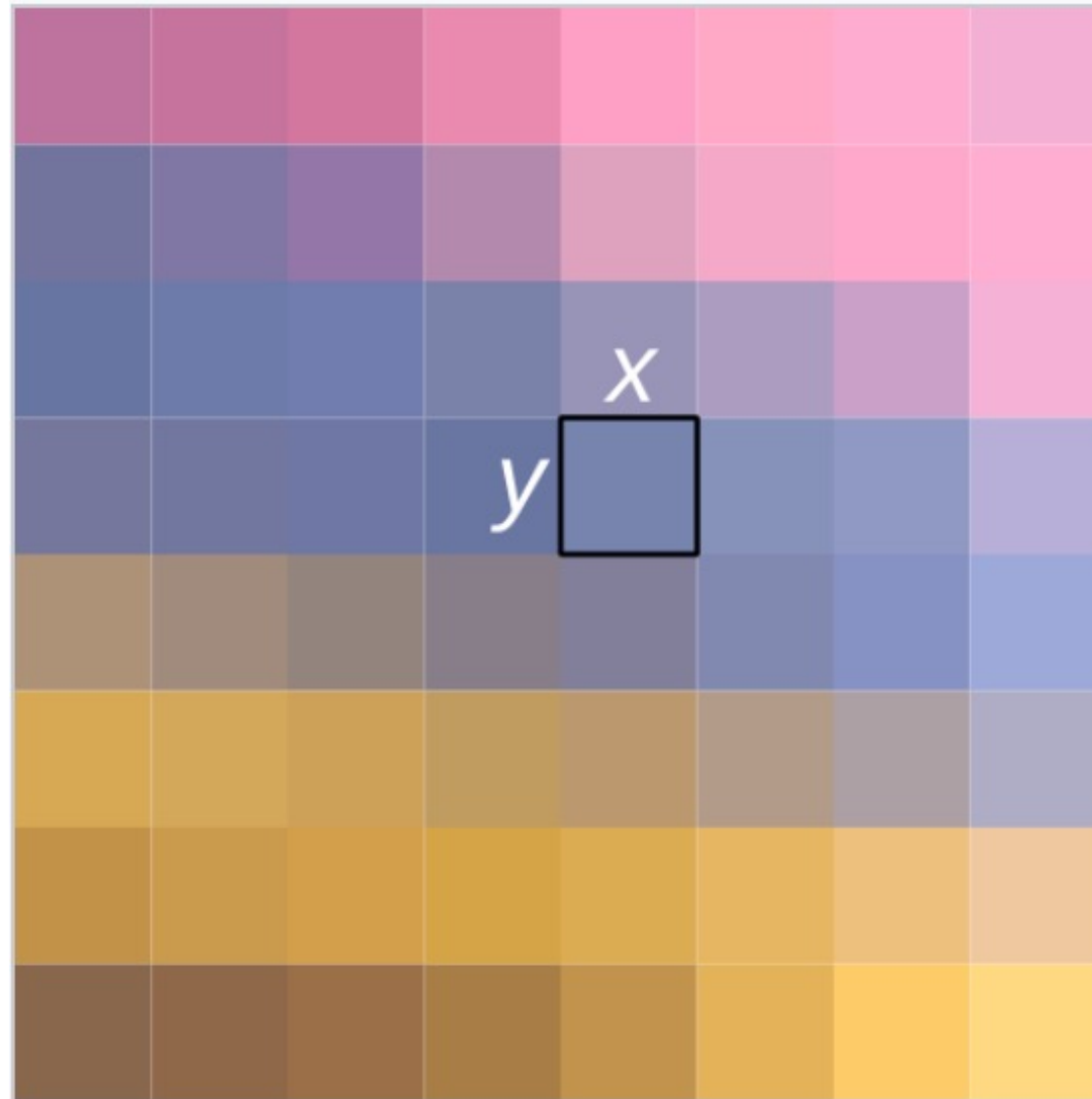
Digital Image

- Imperfections resulting from JPEG quality 90

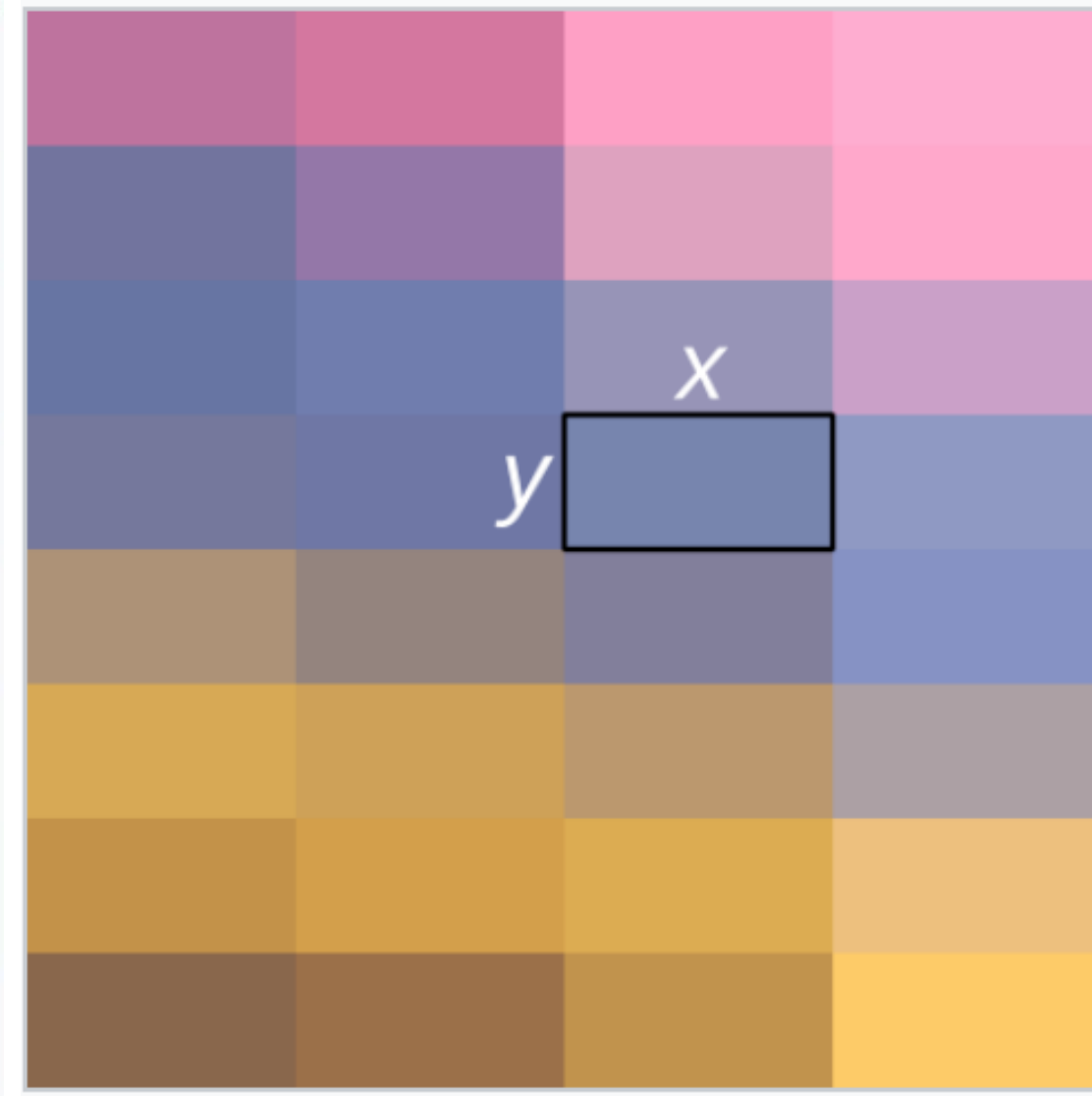


Digital Image

- Pixel aspect ratio
- Pixel size related to physical units: e.g., mm, km
- It corresponds to size of part of physical scene it represents
- **Pixels aren't always square**



Pixel aspect ratio 1:1



Pixel aspect ratio 2:1



Digital Image

- Satellite image
- Physical scale



Digital Image

- Image of same scene (as before) with half the number of pixels along left-right direction
- Effectively, pixel units (scale) are different along left-right and top-down directions
- But display doesn't respect that



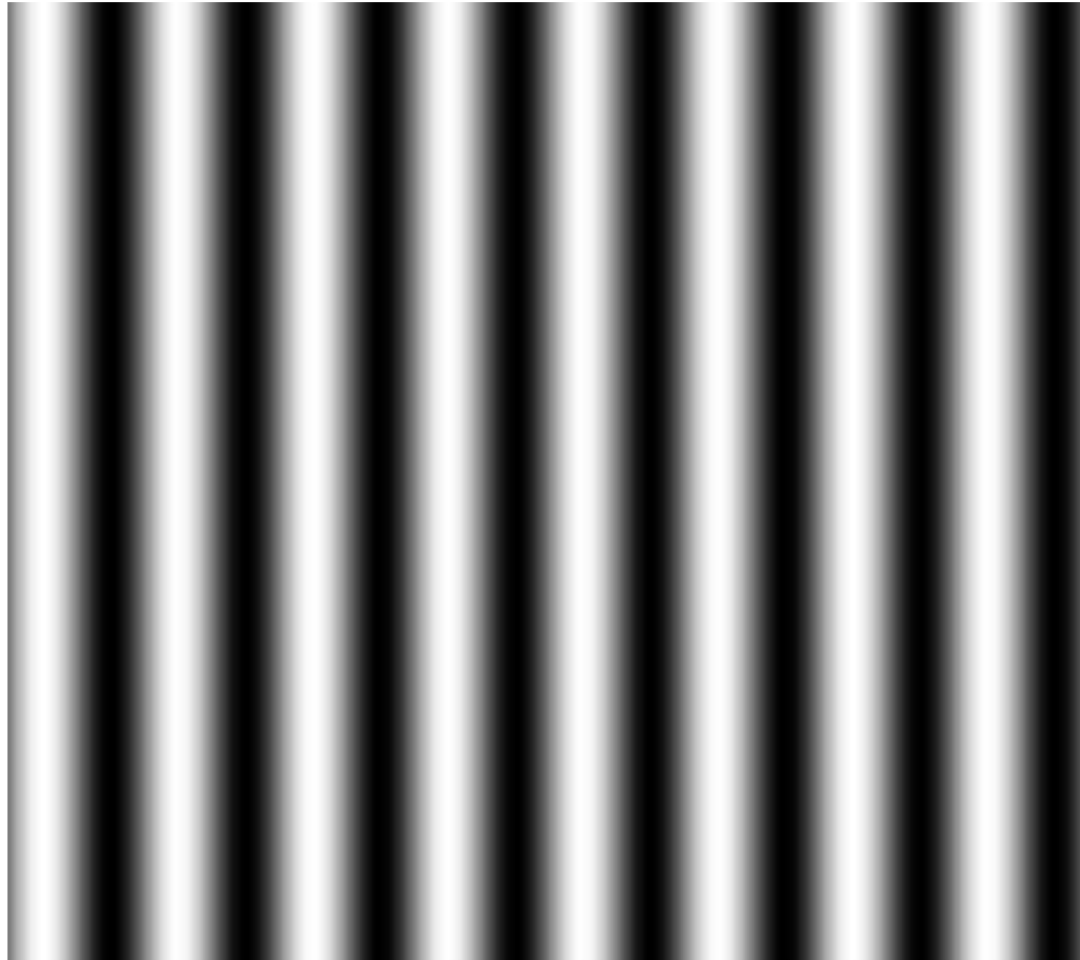
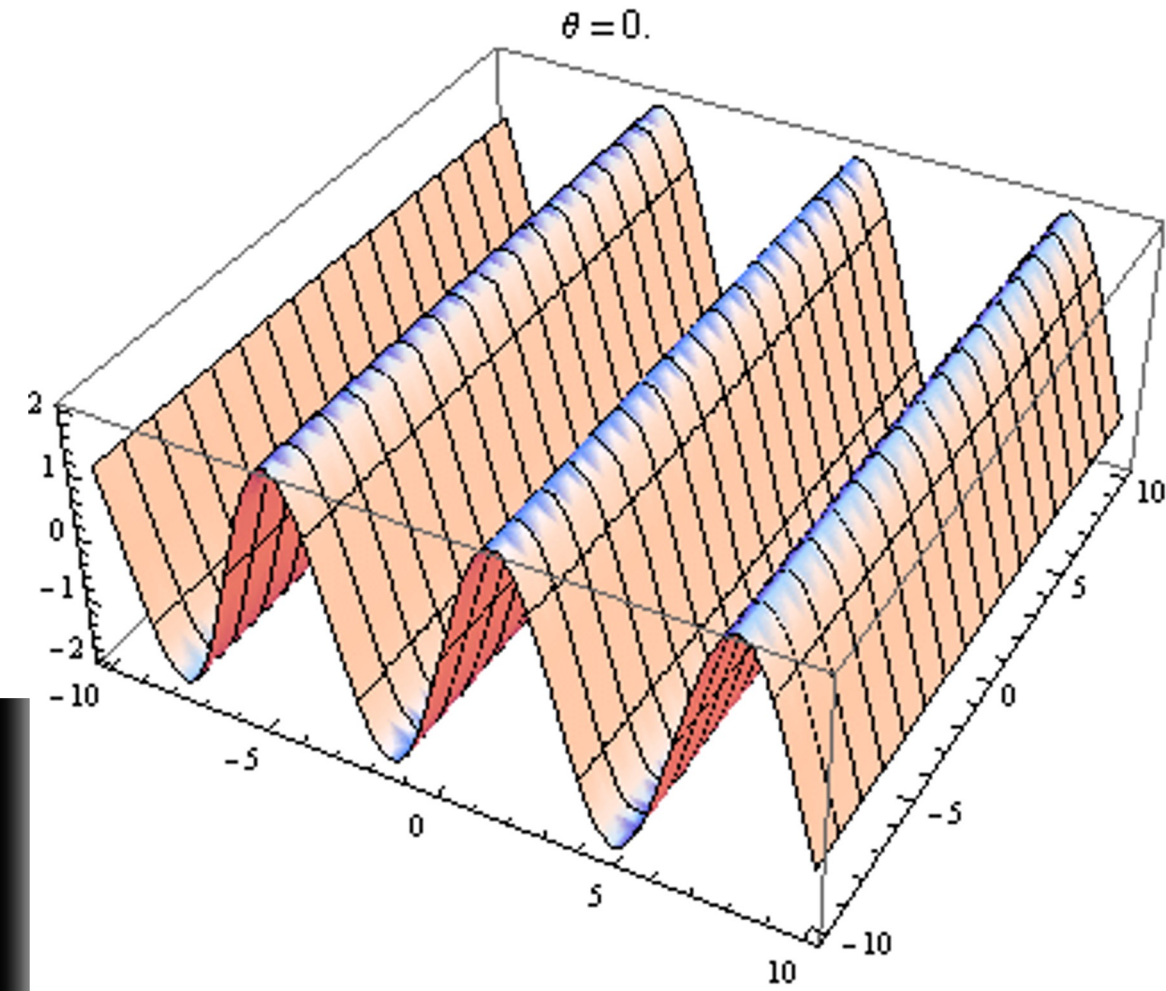
Digital Image

- Resizing bug (during editing)



Digital Image

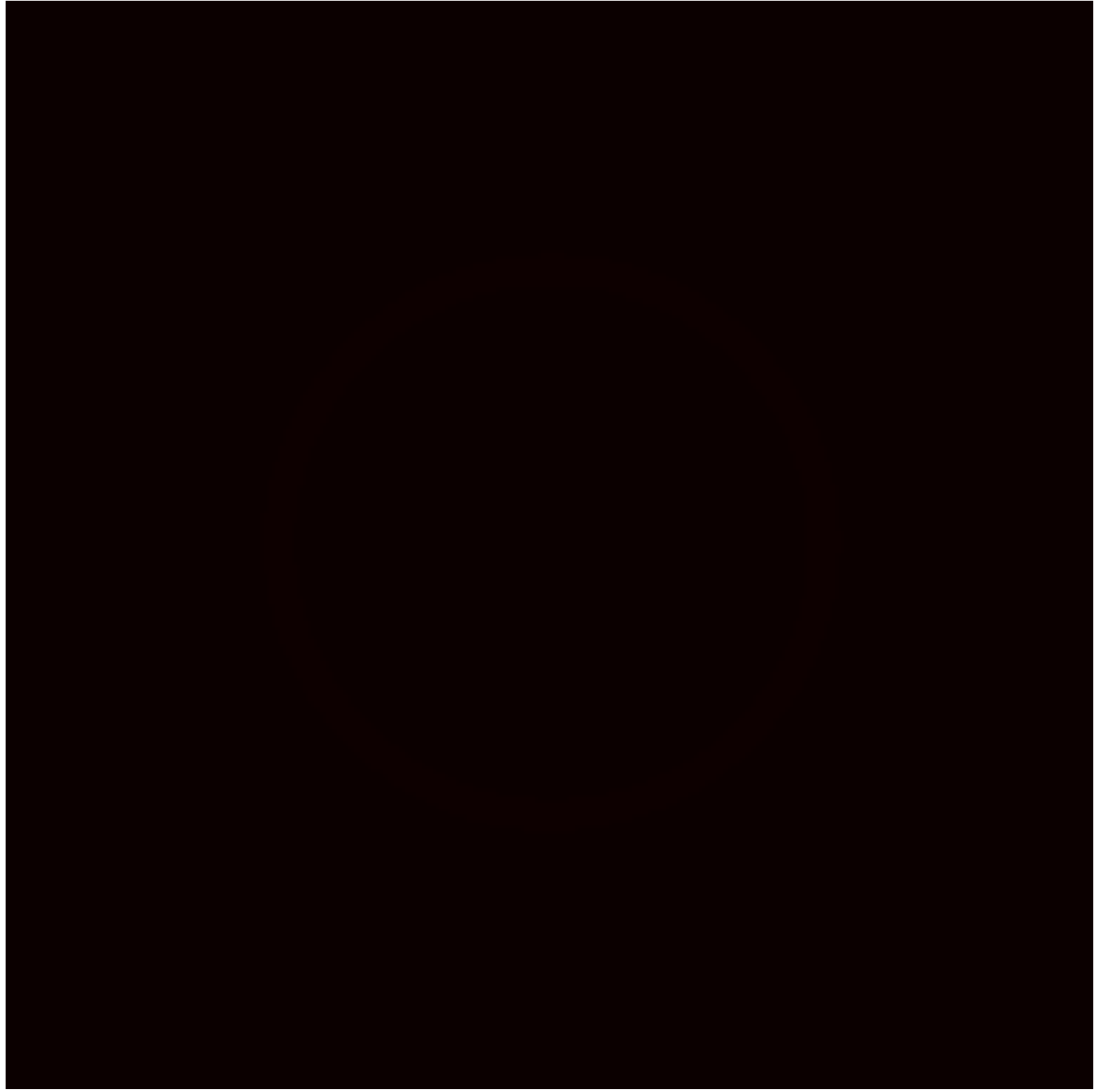
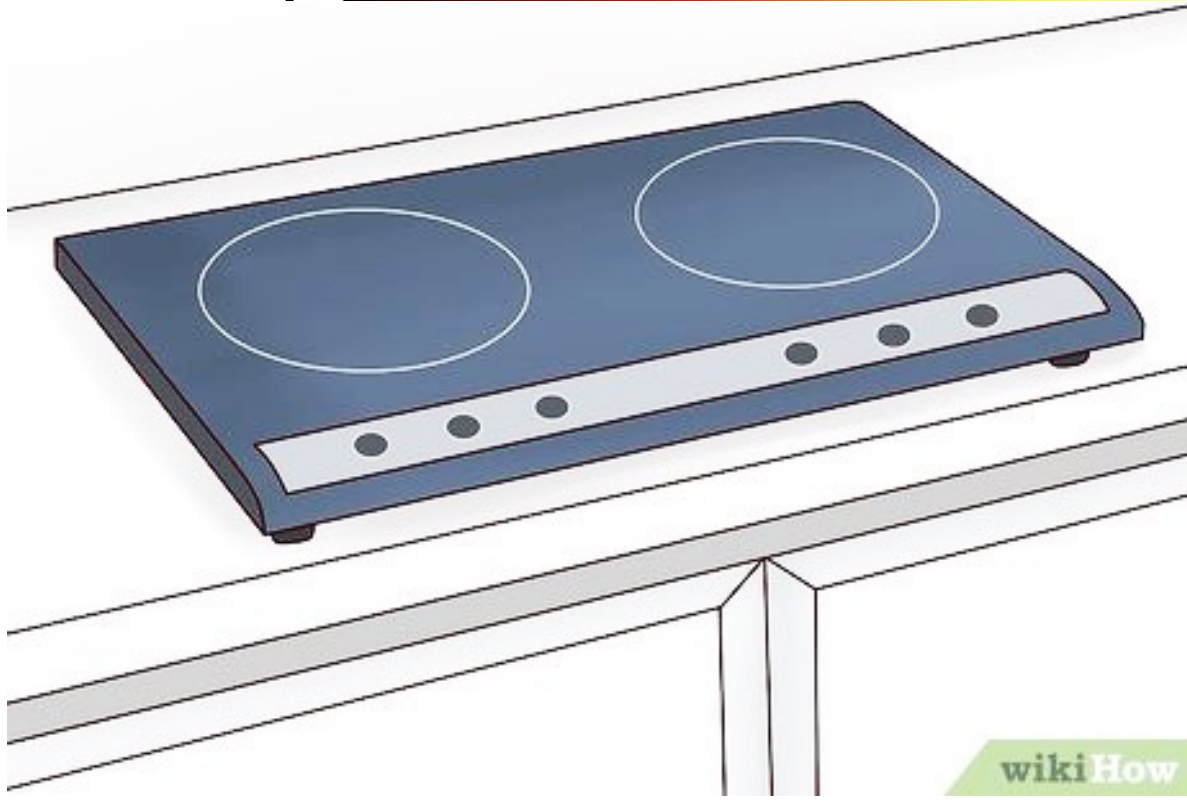
- Image can be modeled as a function
 - e.g., domain: (x,y) in $\mathbb{R}^2$
 - e.g., range: $f(x,y)$ in $\mathbb{R}$
- Sinusoid function



- Sinusoid image

Digital Image

- Consider function $f(x,y,t)$ modeling temperature at time t , at location (x,y) on surface of a stove-top burner
- $f(x,y,t)$ visualized as a video →
- Colormap 



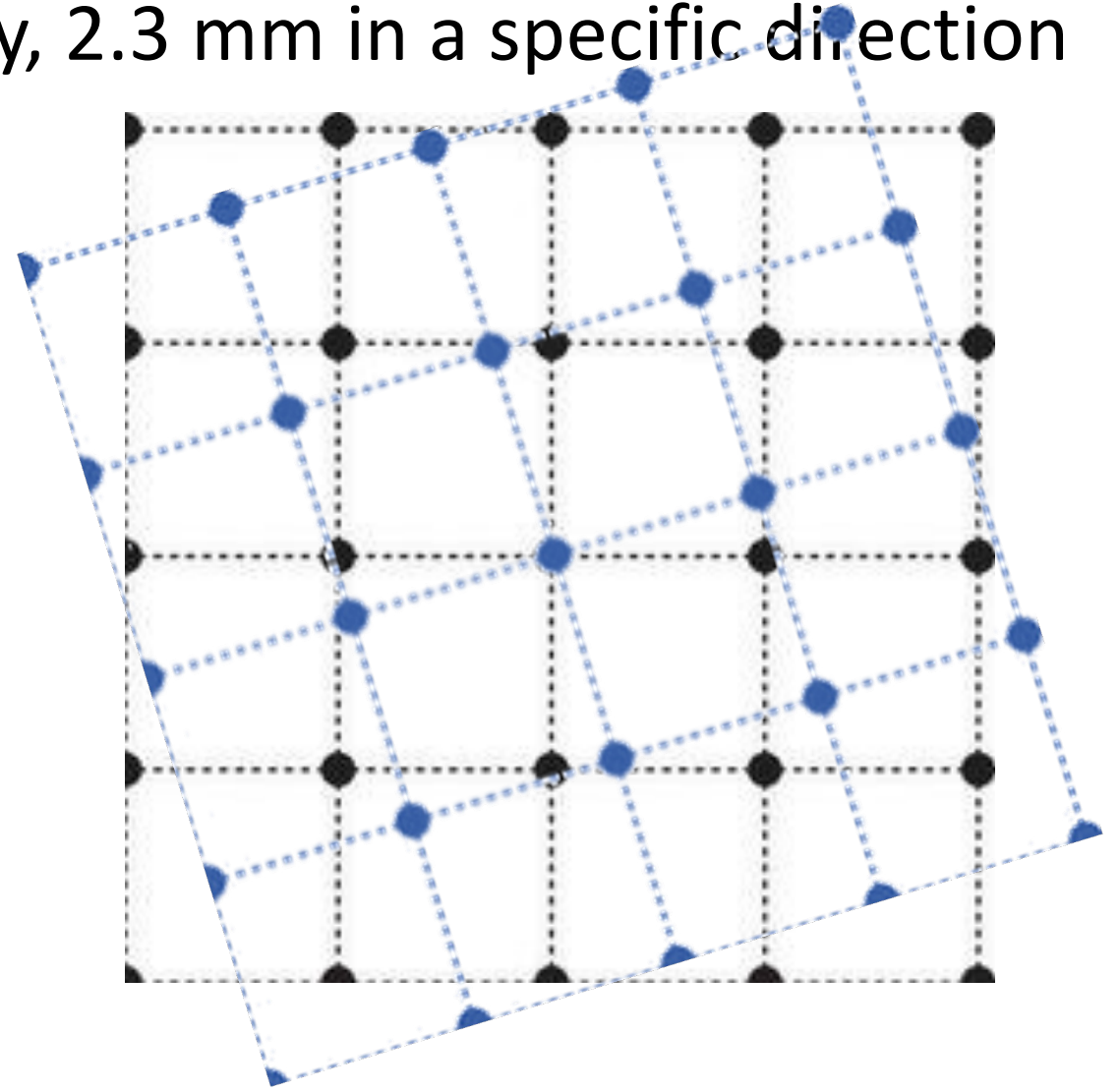
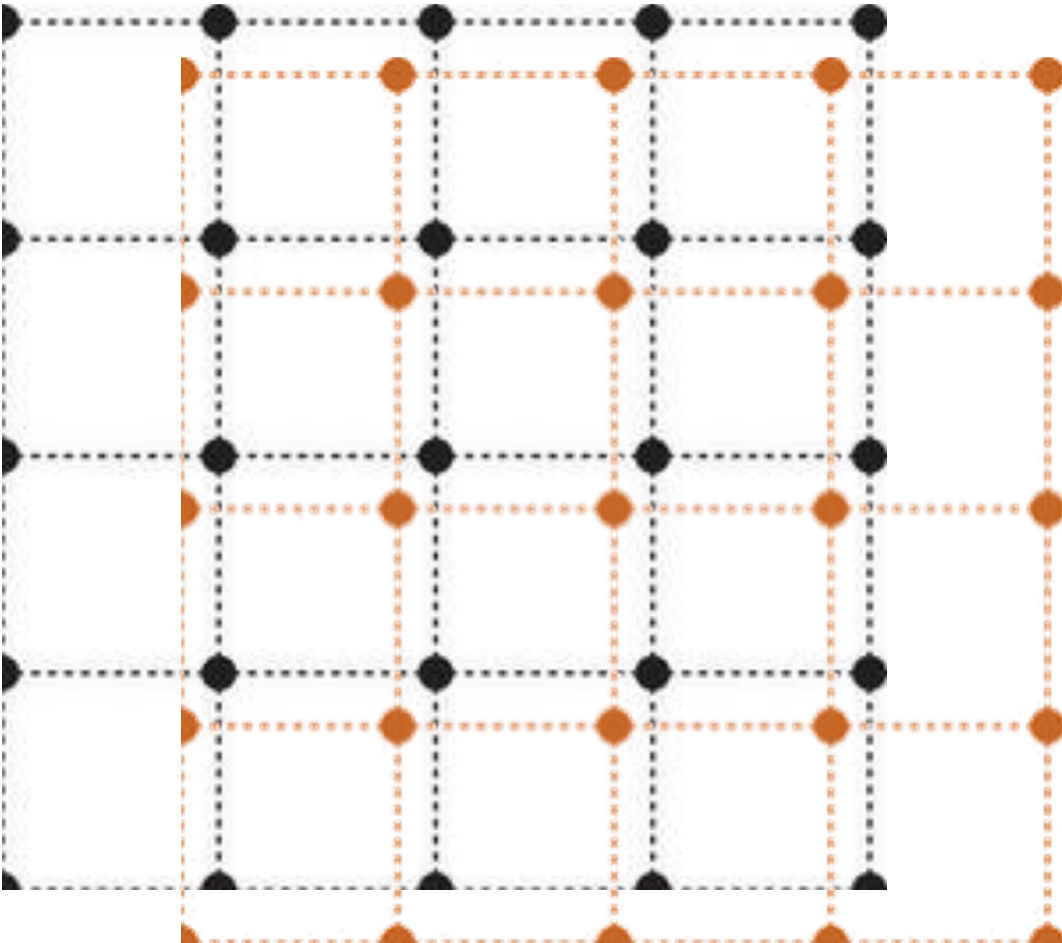
Digital Image Interpolation

- Image **rotation**
- Ratneshwar Mahadev temple
- 9° tilt
- Garbha-griha submerged under Ganga most of the year



Digital Image Interpolation

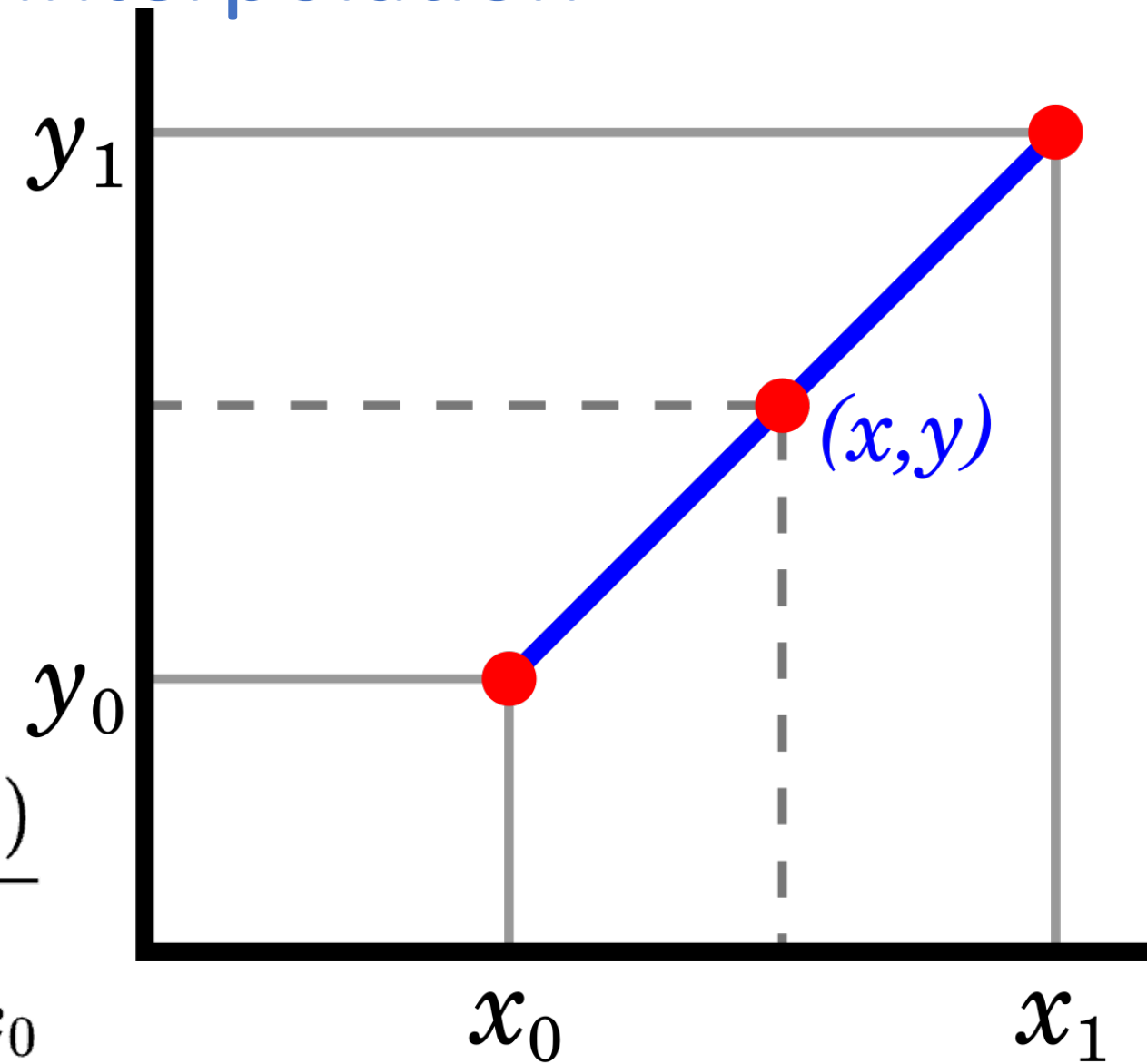
- Interpolating functions
- Motivation
 - Rotate (objects in) image by, say, 4.5 degrees, around a given point
 - Shift/translate (all objects in) image by, say, 2.3 mm in a specific direction



Digital Image Interpolation: Linear Interpolation

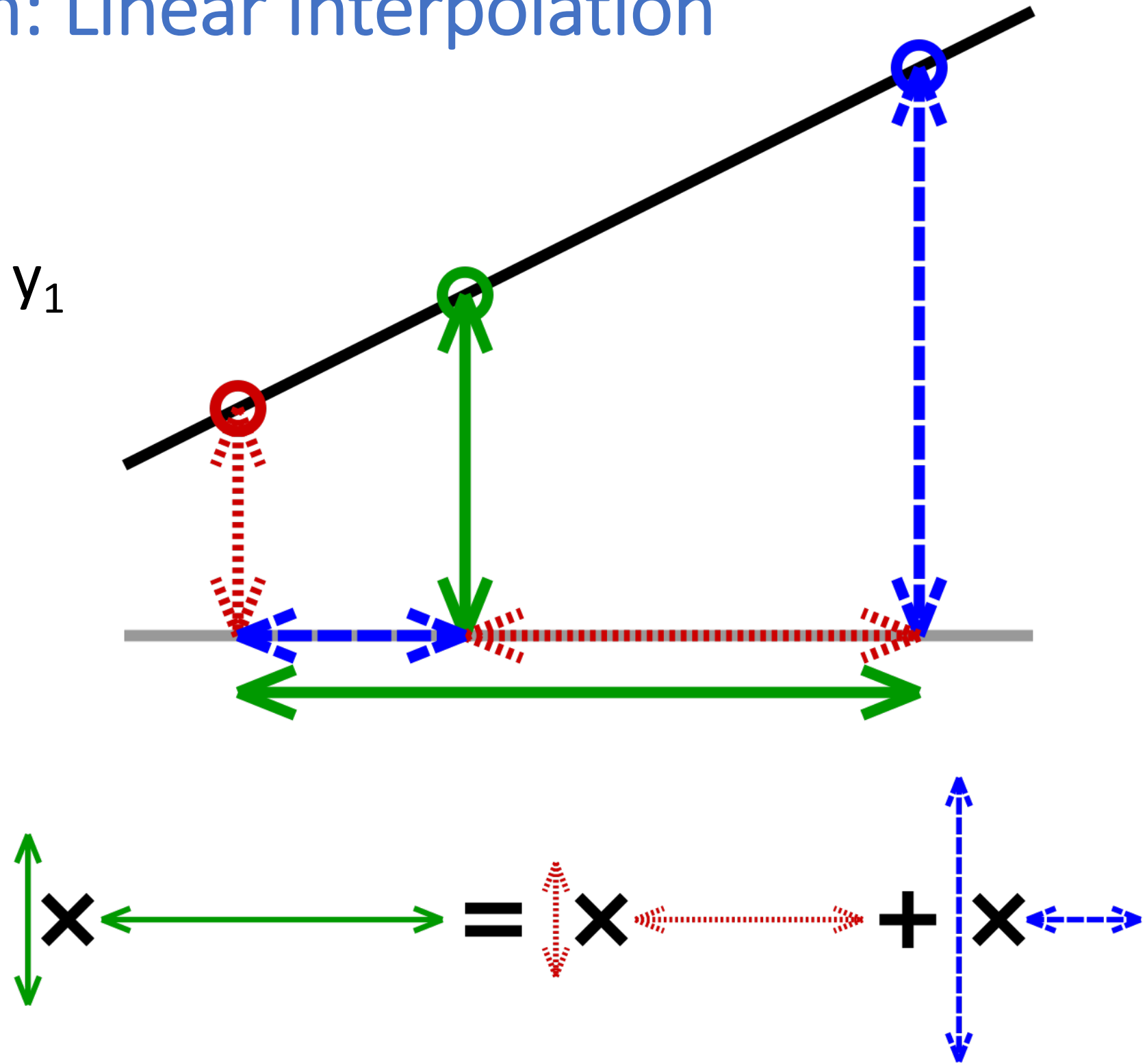
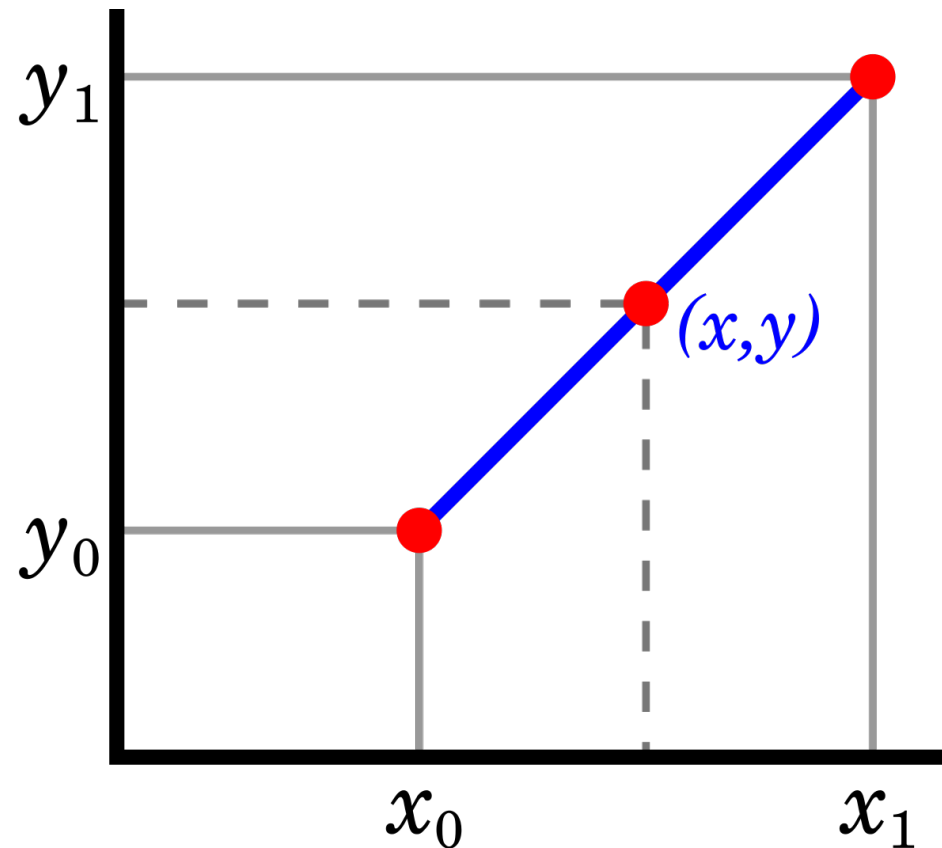
- $y = f(x)$ where x is scalar
- Given: (x_0, y_0) , (x_1, y_1)
- Given: x within interval $[x_0, x_1]$
- Estimate y

$$\begin{aligned} y &= y_0 + (x - x_0) \frac{y_1 - y_0}{x_1 - x_0} \\ &= \frac{y_0(x_1 - x_0)}{x_1 - x_0} + \frac{y_1(x - x_0) - y_0(x - x_0)}{x_1 - x_0} \\ &= \frac{y_1 x - y_1 x_0 - y_0 x + y_0 x_0 + y_0 x_1 - y_0 x_0}{x_1 - x_0} \\ &= \frac{y_0(x_1 - x) + y_1(x - x_0)}{x_1 - x_0}, \end{aligned}$$

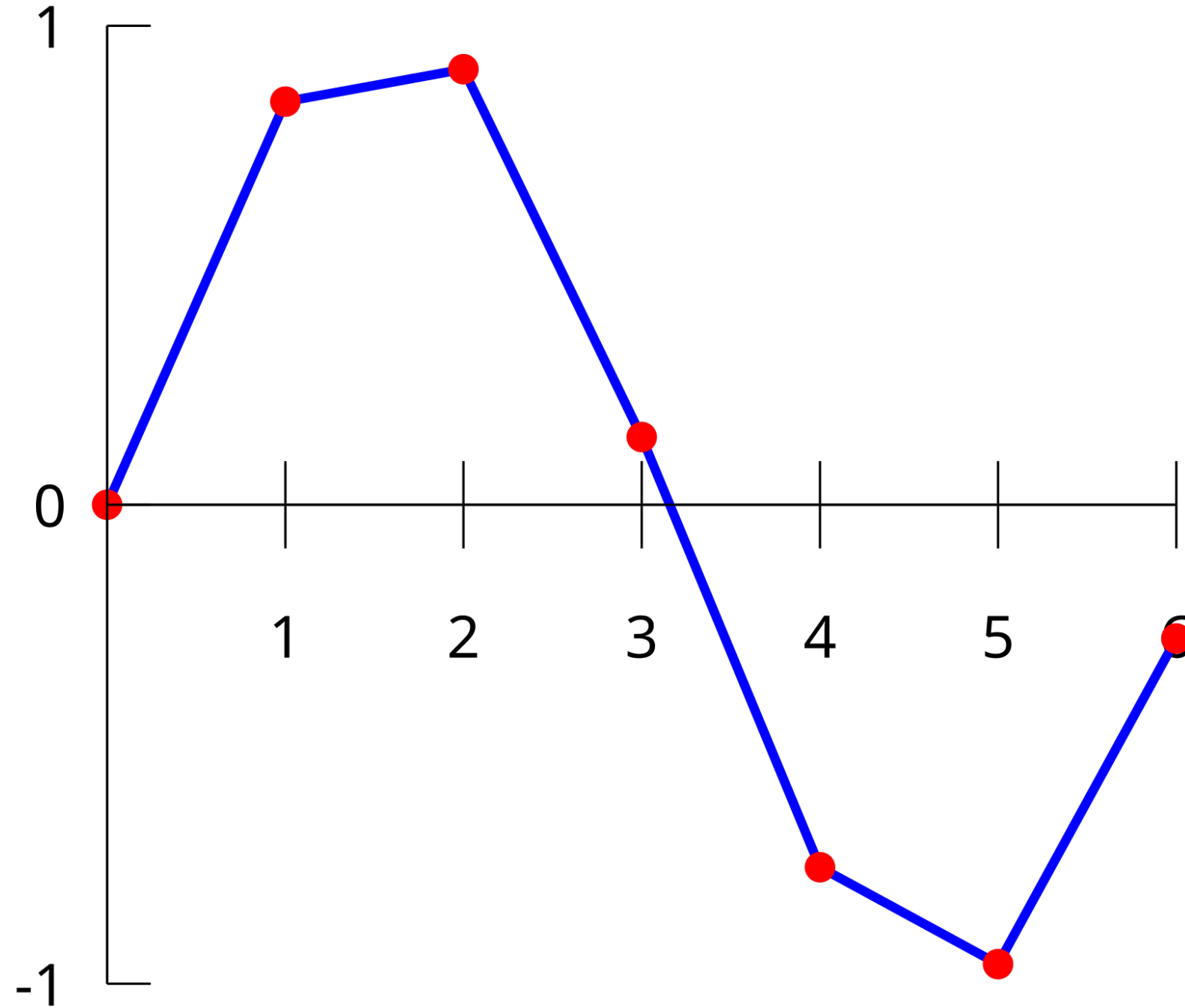


Digital Image Interpolation: Linear Interpolation

- $y = \frac{y_0(x_1 - x) + y_1(x - x_0)}{x_1 - x_0}$
- Convex combination of y_0 and y_1
- y always lies within $[y_0, y_1]$
- Geometric interpretation

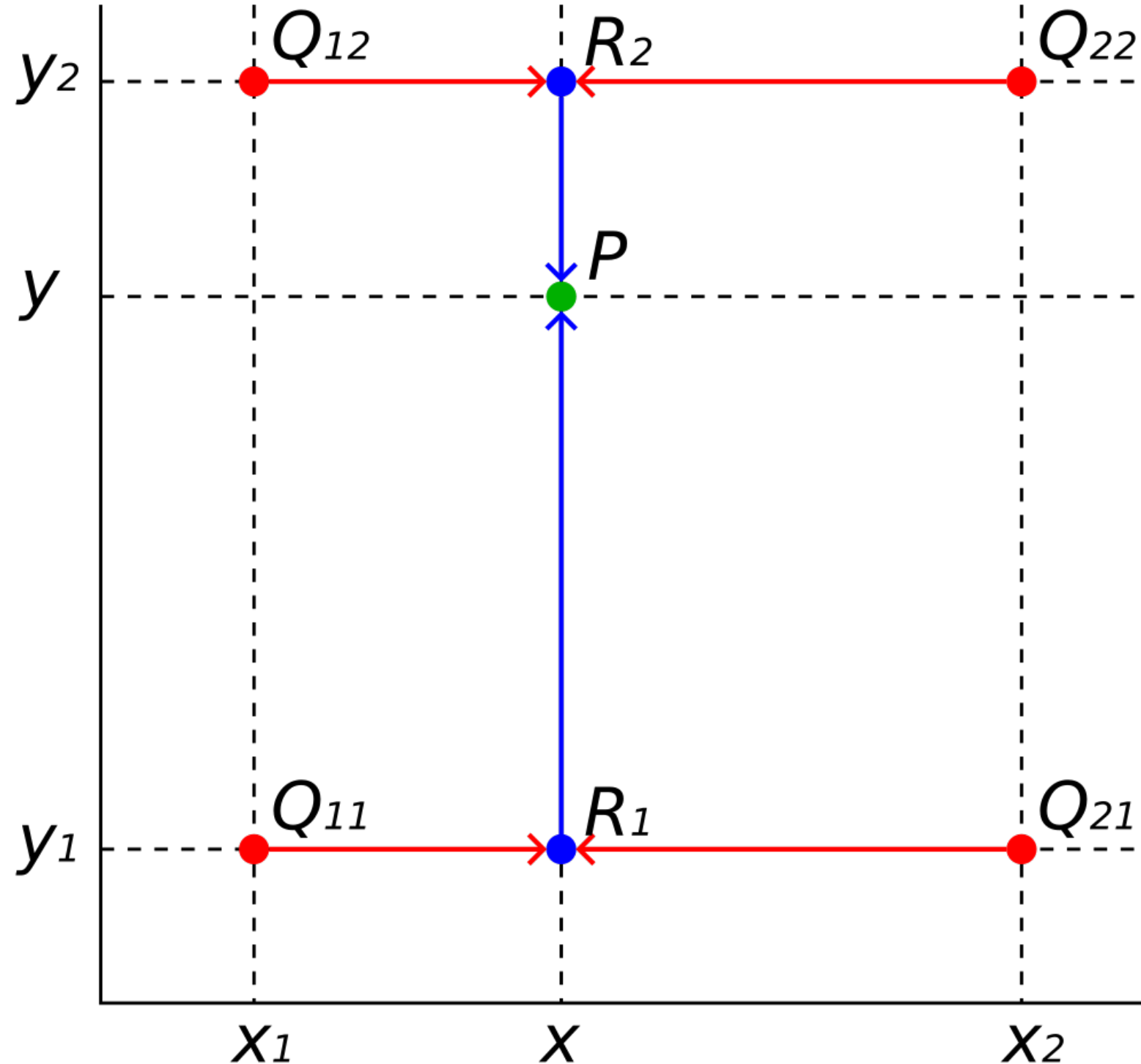


Digital Image Interpolation: Linear Interpolation



Digital Image Interpolation: Bilinear Interpolation

- $z = f(x, y)$
- Locations $Q_{ij} := (x_i, y_j)$
- Given:
 $(Q_{11}, f(Q_{11}))$, $(Q_{12}, f(Q_{12}))$,
 $(Q_{21}, f(Q_{21}))$, $(Q_{22}, f(Q_{22}))$
- Given:
location $P := (x, y)$ where
 x within interval $[x_1, x_2]$,
 y within interval $[y_1, y_2]$
- Estimate:
value $f(P)$



Digital Image Interpolation: Bilinear Interpolation

- First, linearly interpolate along x axis

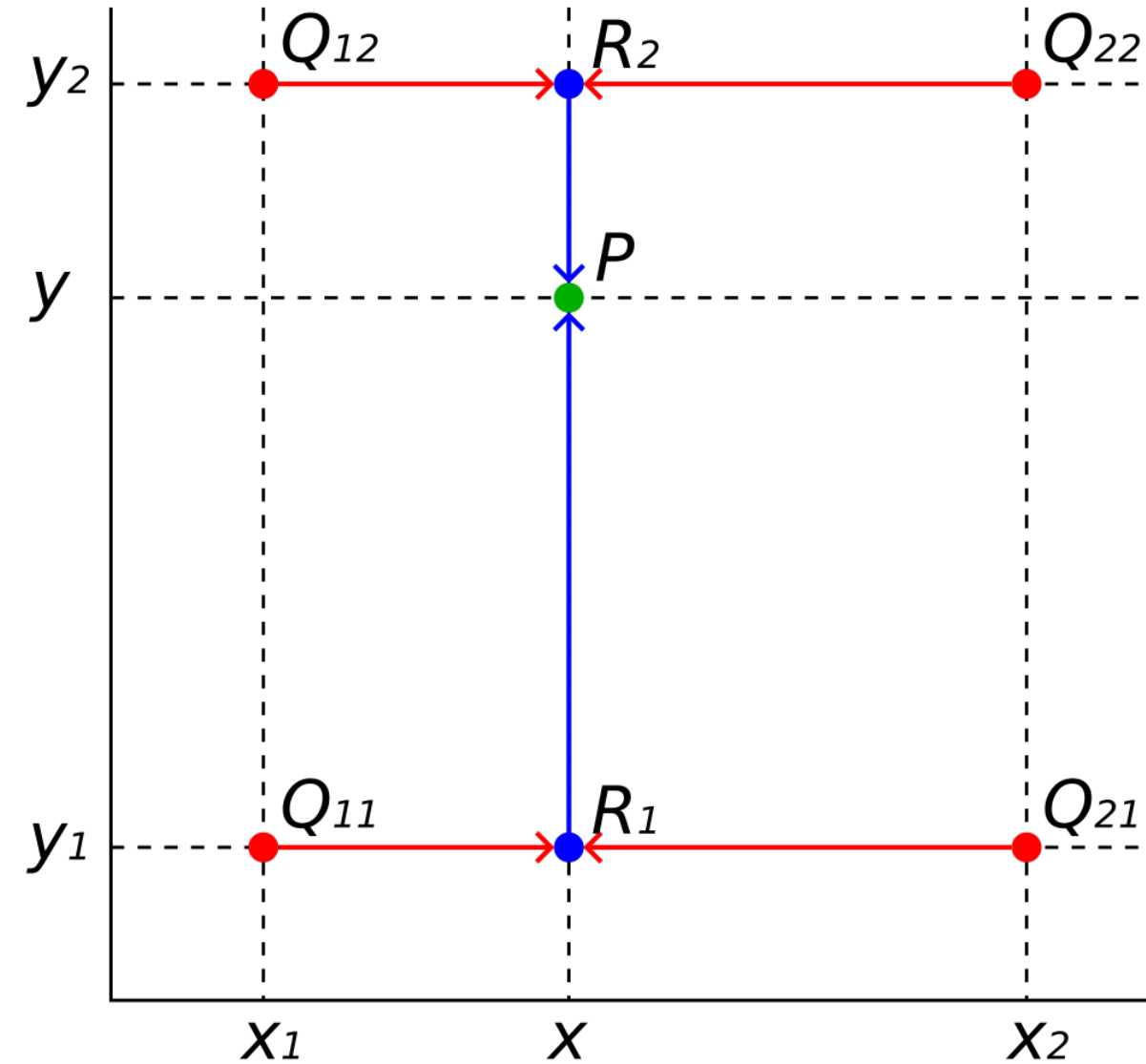
$$f(x, y_1) = \frac{x_2 - x}{x_2 - x_1} f(Q_{11}) + \frac{x - x_1}{x_2 - x_1} f(Q_{21})$$

$$f(x, y_2) = \frac{x_2 - x}{x_2 - x_1} f(Q_{12}) + \frac{x - x_1}{x_2 - x_1} f(Q_{22})$$

- Then, linearly interpolate along y axis

$$f(x, y) = \frac{y_2 - y}{y_2 - y_1} f(x, y_1) + \frac{y - y_1}{y_2 - y_1} f(x, y_2)$$

$$\begin{aligned} &= \frac{1}{(x_2 - x_1)(y_2 - y_1)} (f(Q_{11})(x_2 - x)(y_2 - y) + f(Q_{12})(x_2 - x)(y - y_1) \\ &\quad + f(Q_{21})(x - x_1)(y_2 - y) + f(Q_{22})(x - x_1)(y - y_1)) \end{aligned}$$

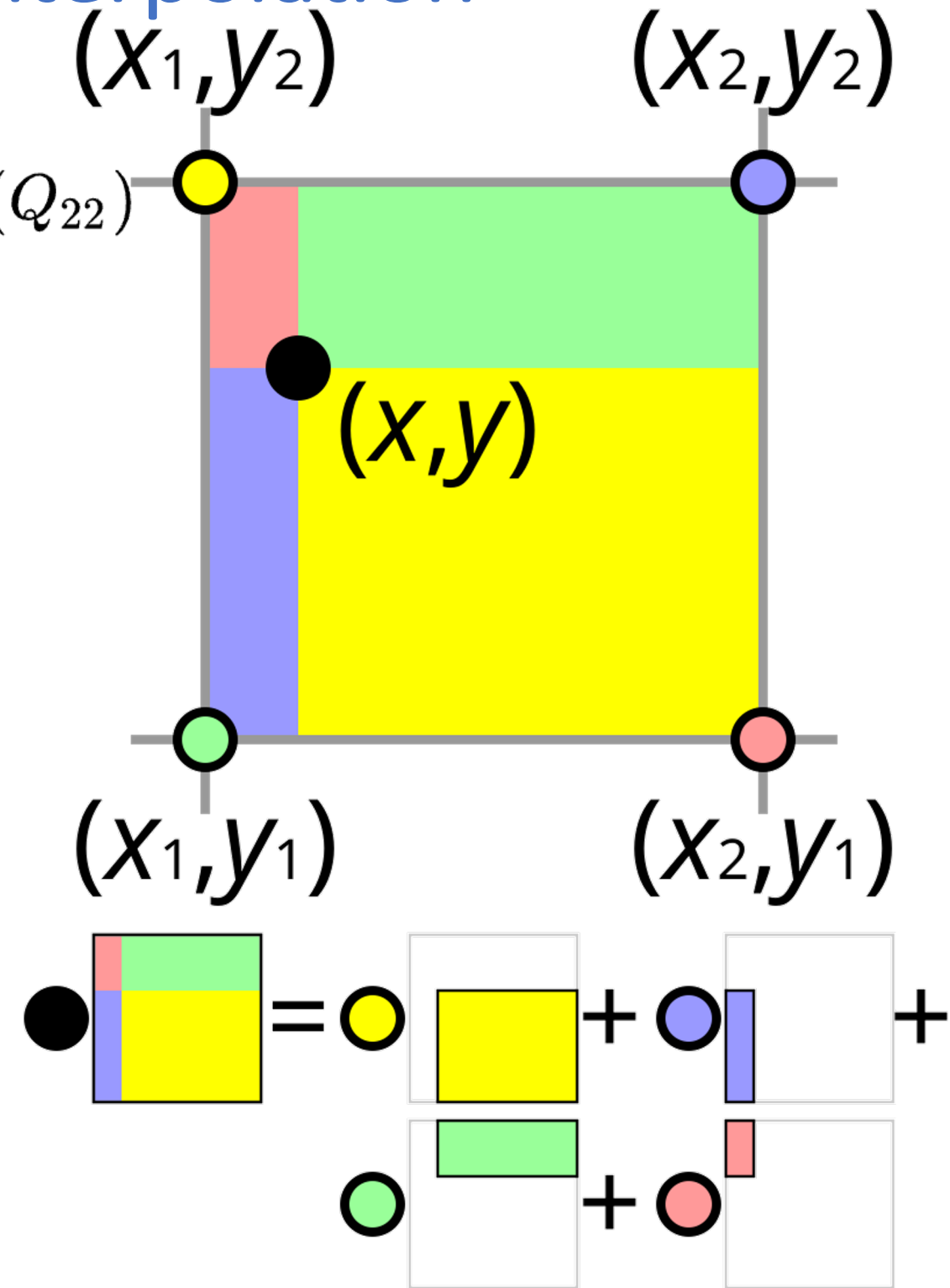


Digital Image Interpolation: Bilinear Interpolation

- So

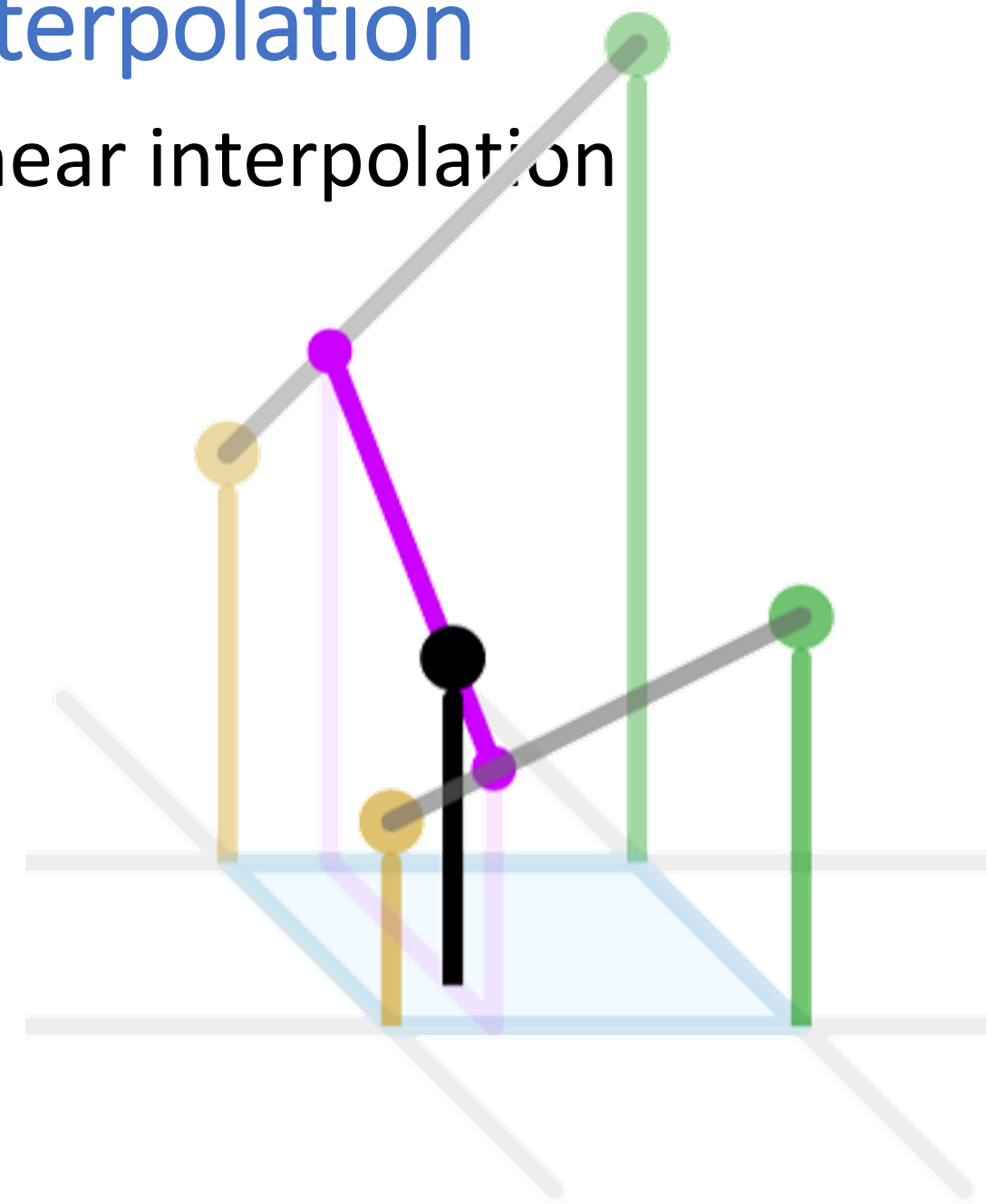
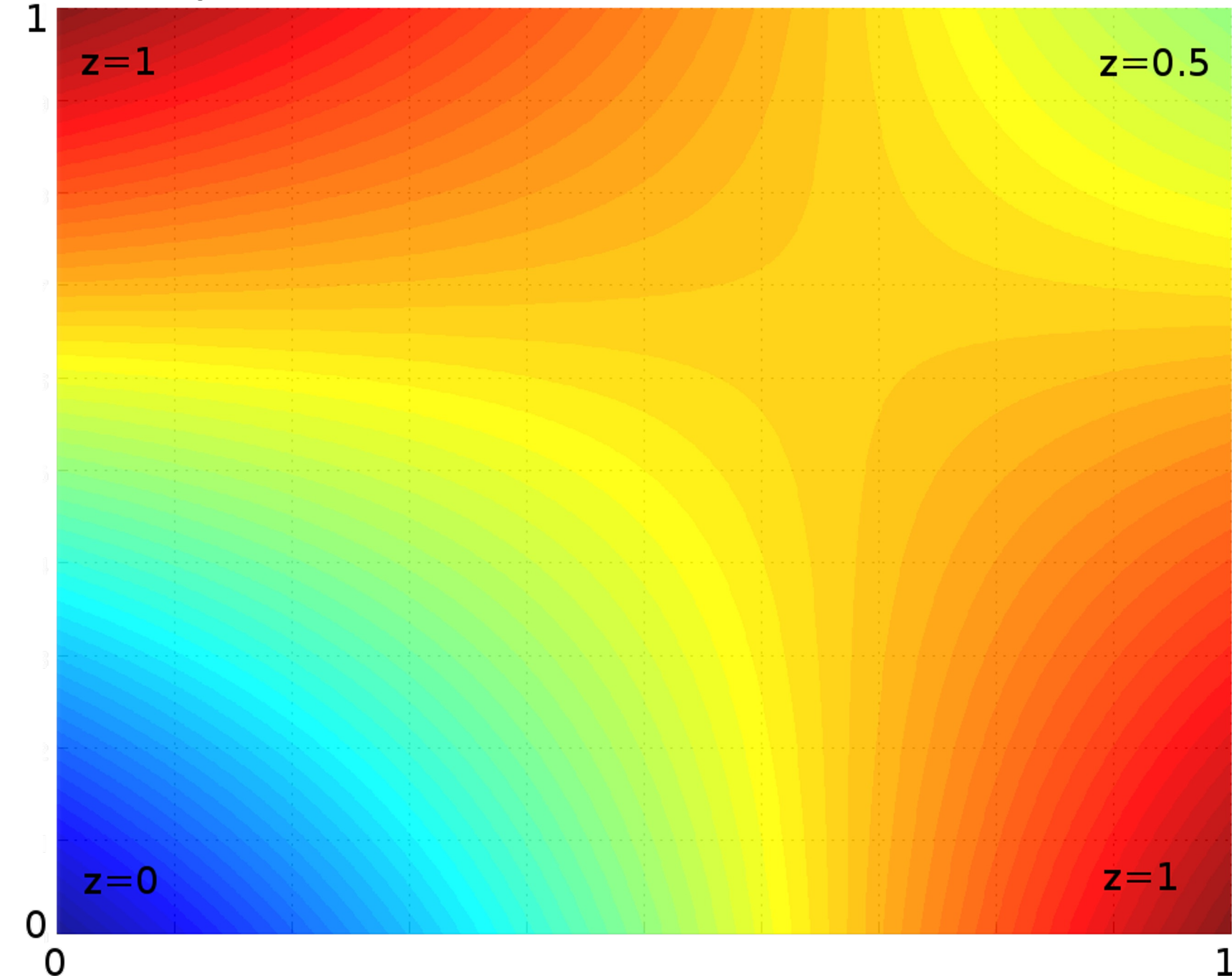
$$f(x, y) \approx w_{11} f(Q_{11}) + w_{12} f(Q_{12}) + w_{21} f(Q_{21}) + w_{22} f(Q_{22})$$

- which is a convex combination of $f(Q_{11})$, $f(Q_{12})$, $f(Q_{21})$, $f(Q_{22})$
- $f(x, y)$ guaranteed to lie within $(\min[f(Q_{11}), f(Q_{12}), f(Q_{21}), f(Q_{22})], \max[f(Q_{11}), f(Q_{12}), f(Q_{21}), f(Q_{22})])$
- Geometric interpretation
- Independent of order of interpolation steps along X and Y axes



Digital Image Interpolation: Bilinear Interpolation

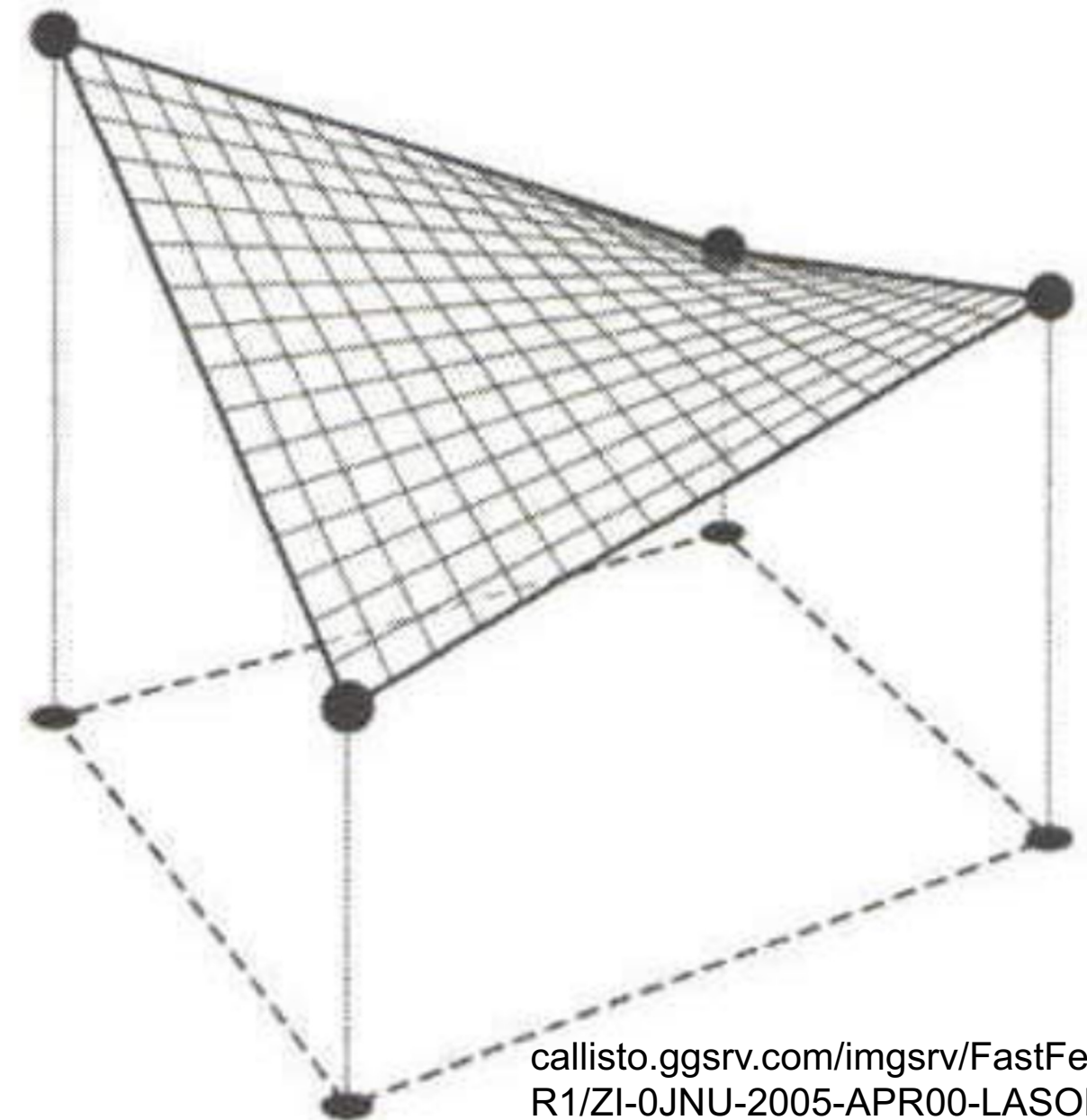
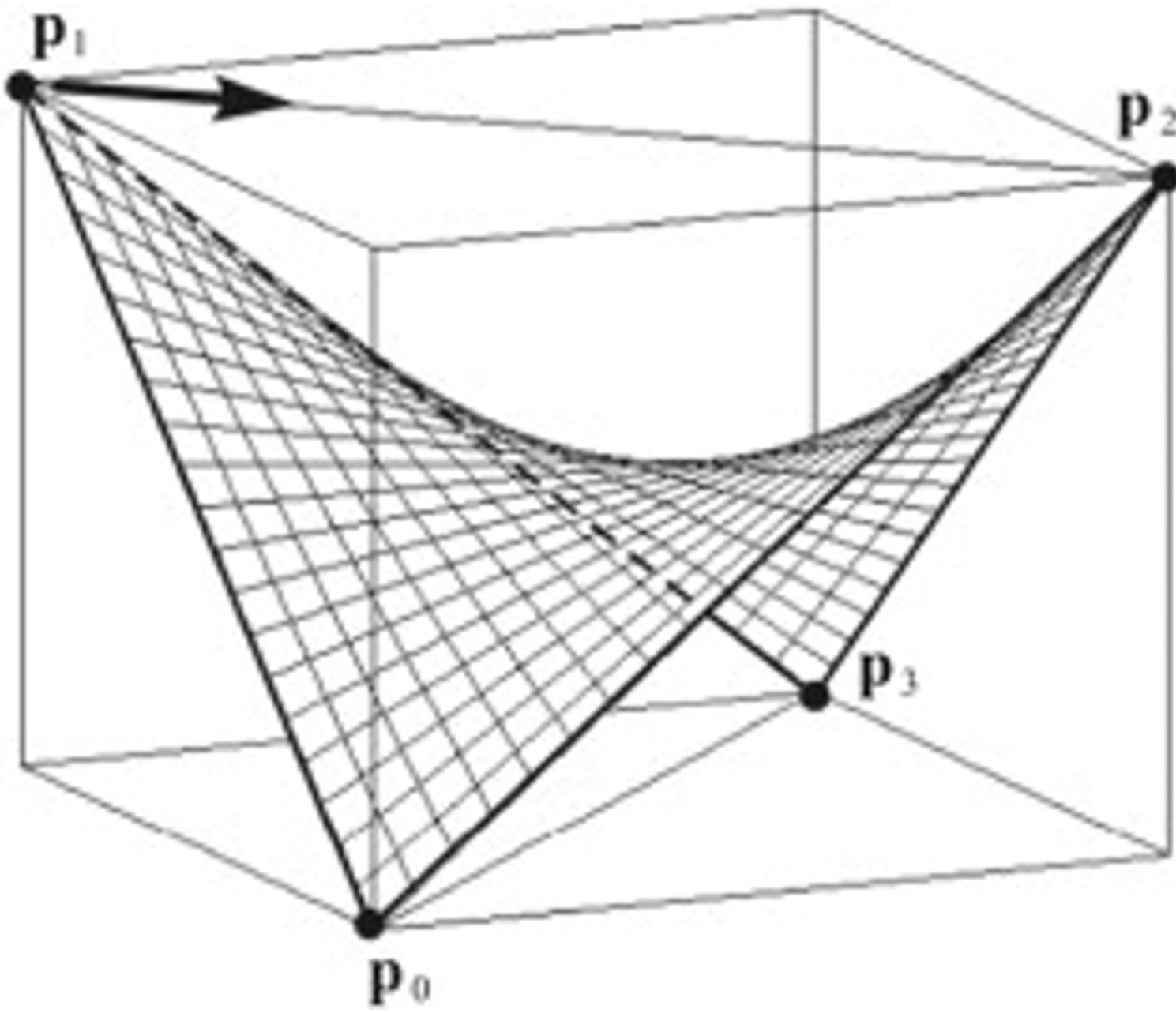
- Interpolated surface can be curved, unlike linear interpolation



Bilinear

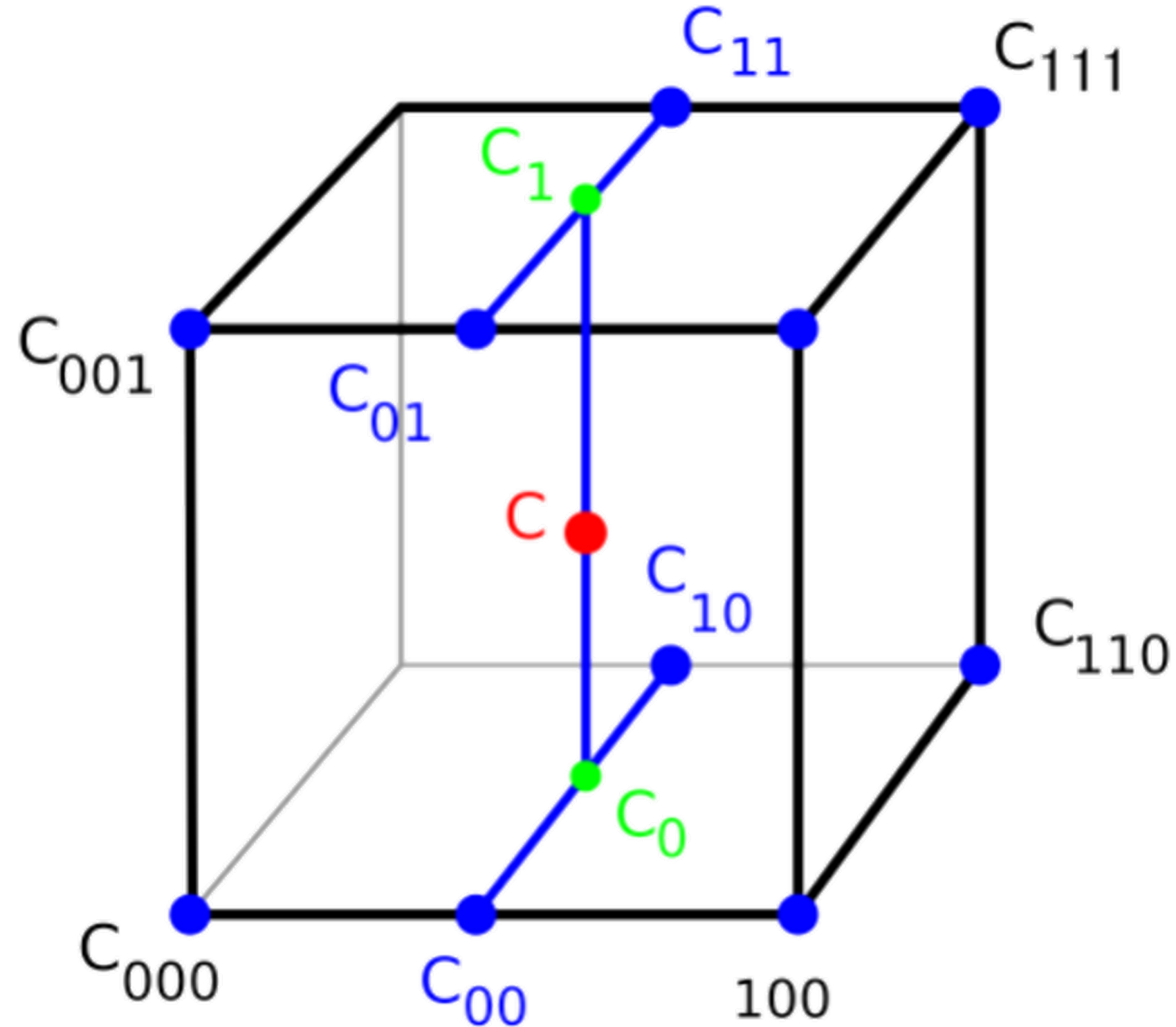
Digital Image Interpolation: Bilinear Interpolation

- Interpolated surface is curved, unlike linear interpolation



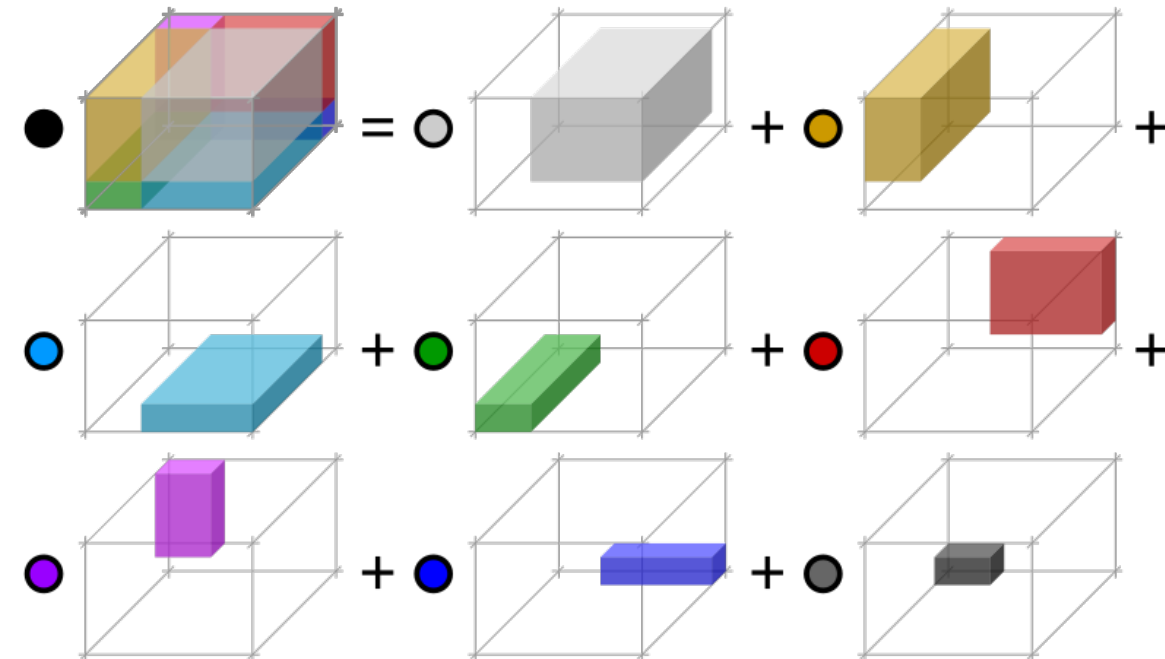
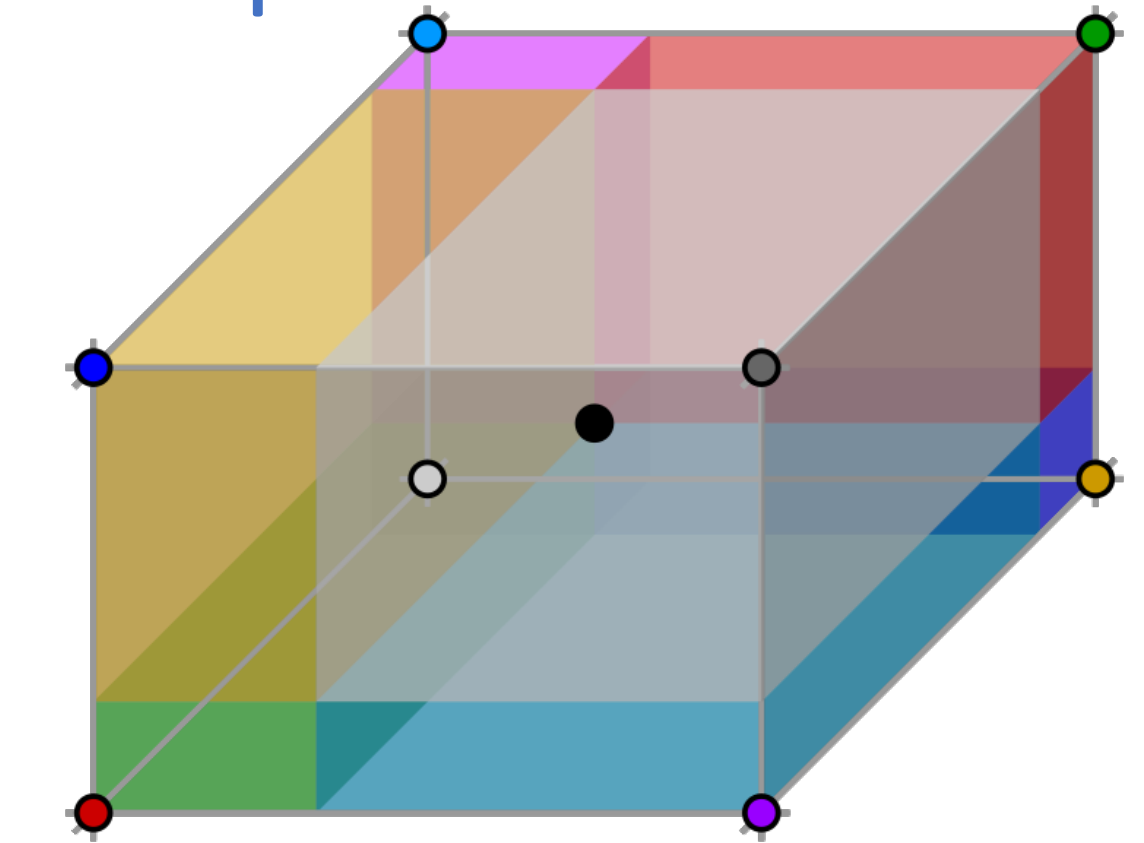
Digital Image Interpolation: Trilinear Interpolation

- Trilinear interpolation
- 2 bilinear interpolations followed by 1 linear interpolation
- Independent of order of interpolation steps along 3 axes



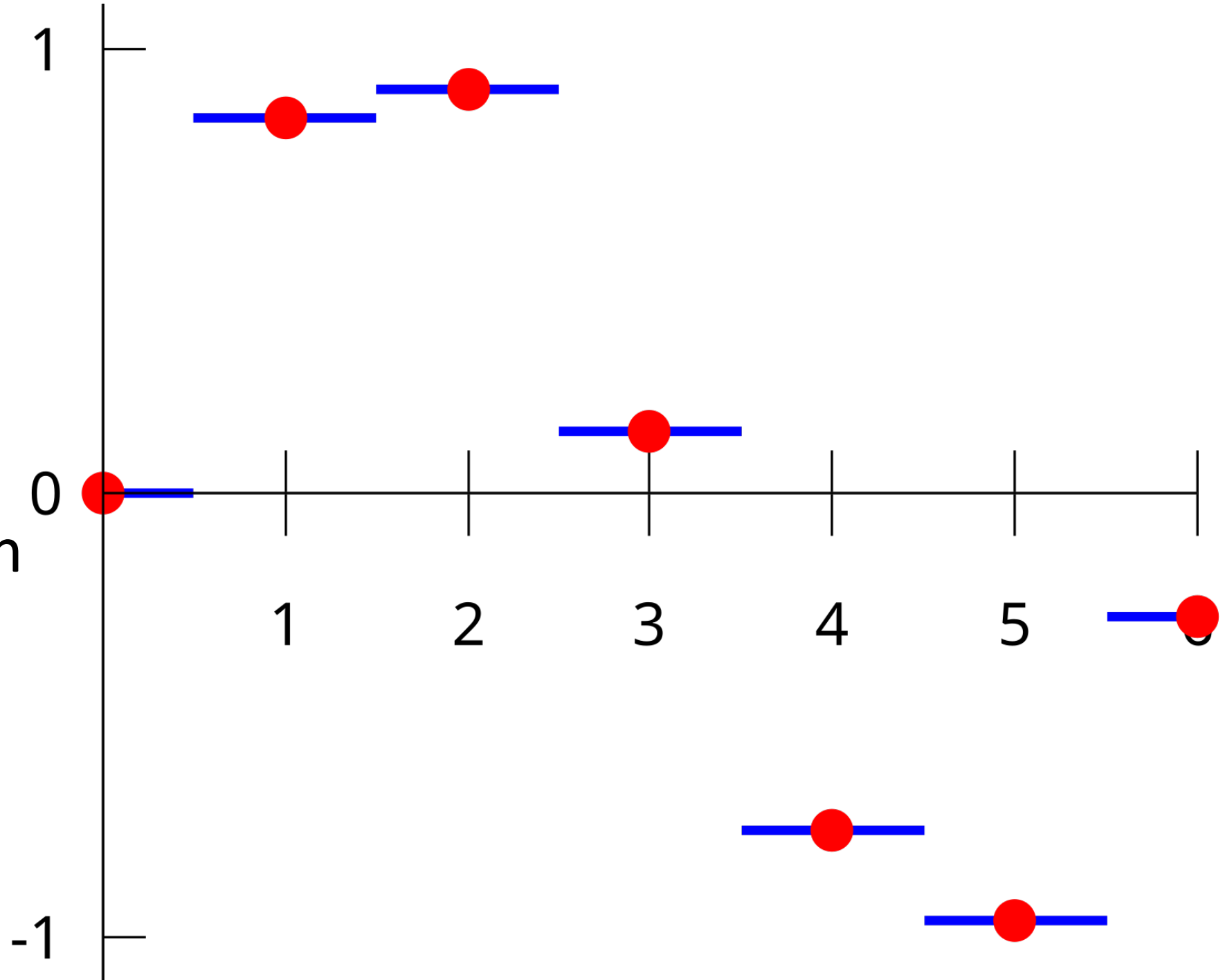
Digital Image Interpolation: Trilinear Interpolation

- Geometric interpretation
- Value at desired point *
entire volume
=
summation over all corners *
value at each corner *
partial volume diagonally opposite corner



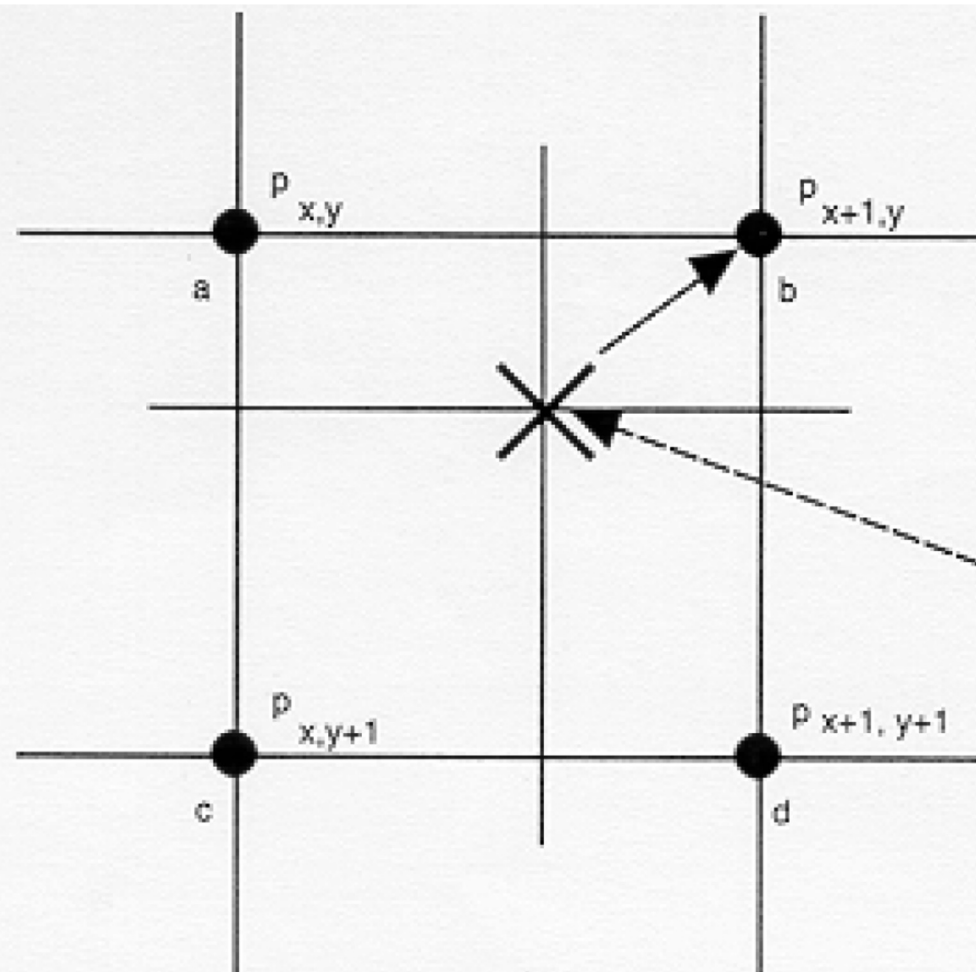
Digital Image Interpolation: Nearest-Neighbor

- For scalar variable x
- Given:
 $(x, f(x))$ values in red
- Estimates:
 $(y, f(y))$ values in blue
- Estimated continuous-domain function is piecewise constant

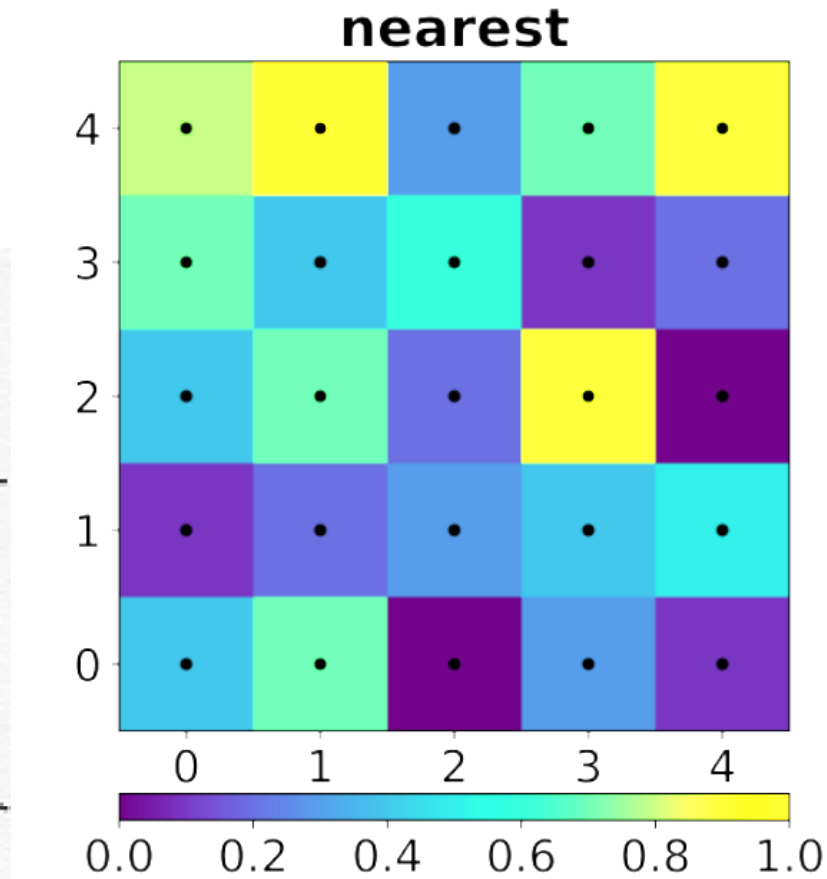


Digital Image Interpolation: Nearest-Neighbor

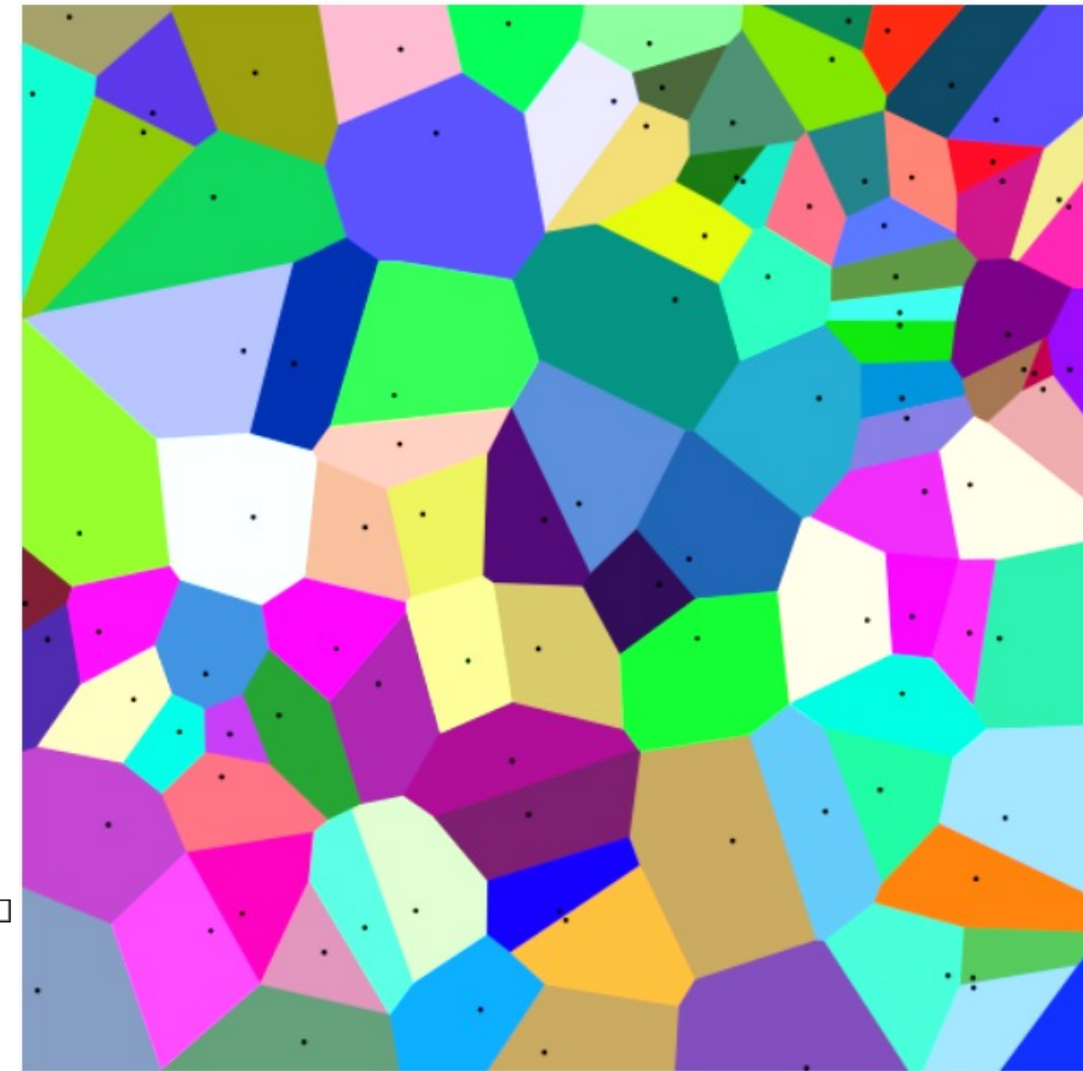
- Nearest-neighbor interpolation in 2D



- Also works on unstructured grid



Nearest neighbor interpolation on a uniform 2D grid (black points). Each colored cell indicates the area in which all the points have the black point in the cell as their nearest black point.



This [Voronoi diagram](#) is an example of nearest neighbor interpolation of a random set of points (black dots) in 2D.

Georgy Voronoy

- Russian mathematician of Ukrainian descent
- At age 40, Voronoy began having stomach problems
 - “Yesterday I had for the first time a clear idea of the algorithm in the theory of forms I am investigating, but also suffered a strong attack of bilious colic which prevented me from working in the evening and from sleeping the whole night. I am so afraid that the results of my enduring efforts, obtained with such difficulty, will perish along with me.”
- Died that year
- Had great influence on Boris Delaunay → Delaunay triangulation
 - Dual of Voronoi diagram
- Son Yurii Voronyi, Ukrainian surgeon, performed world's first human-to-human kidney transplant in 1933

Digital Image Interpolation

- Interpolation of small images



- en.wikipedia.org/wiki/Comparison_gallery_of_image_scaling_algorithms#

- Original large/high-res.



- Nearest neighbor.

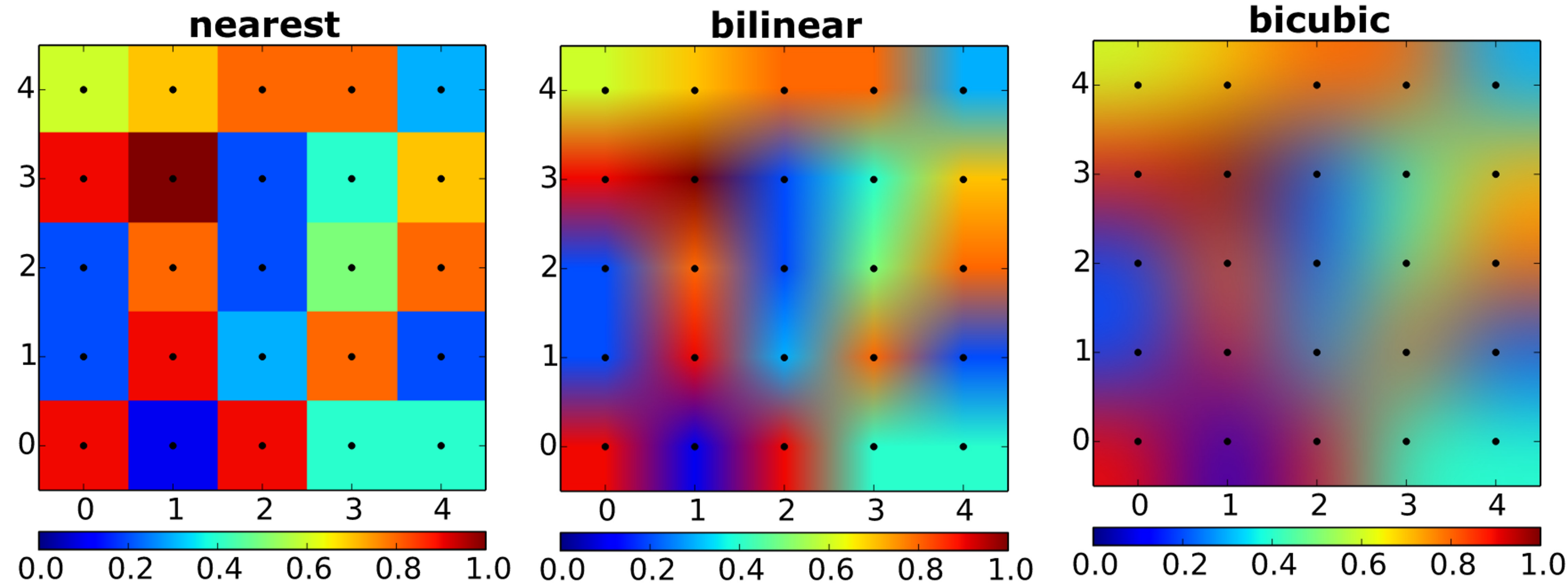


- Bilinear.



Digital Image Interpolation

- Interpolation
 - Bilinear interpolation gives smoother function estimate than nearest neighbor
 - Can we use higher-order polynomials within each piece ? Yes, e.g., bicubic



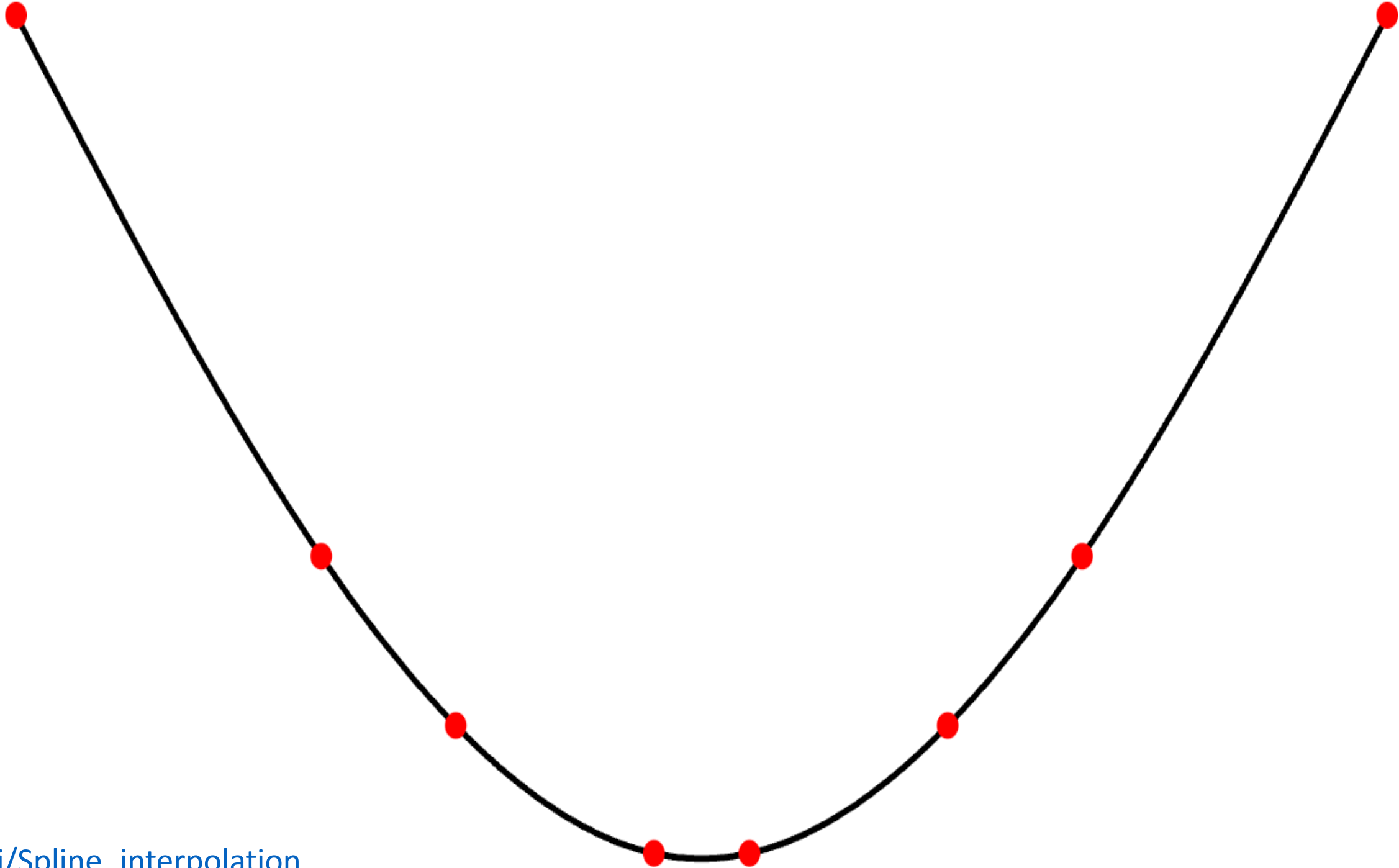
Digital Image Interpolation: Spline Interpolation

- **Spline** (English) =
“a long, thin piece of wood or metal used in building”
 - dictionary.cambridge.org/dictionary/english/spline
- Term “spline” comes from flexible spline devices used by shipbuilders & draftsmen to draw smooth shapes passing via predefined points/“knots”



Digital Image Interpolation: Spline Interpolation

- In our context:
spline =
function
defined
piecewise
by
polynomials



Digital Image Interpolation: Quadratic Spline

- 1D case. **N intervals. Quadratic** polynomial on **each interval**.
- Each quadratic ($a_i + b_i x + c_i x^2$) has 3 parameters $\rightarrow$ total $3N$ unknowns
- **Constraints: within each interval**
 - Value of quadratic must match data at left end $\rightarrow N$ constraints
 - Value of quadratic must match data at right end $\rightarrow N$ constraints
- **Constraints: at joints of intervals (total $N-1$ joints)**
 - First derivative of left & right quadratic must match $\rightarrow N-1$ constraints
- **Constraints: boundary conditions**
 - e.g., assume linearity for either first spline or last spline $\rightarrow 1$ constraint

Digital Image Interpolation: Quadratic Spline

- Data: $(x_1, y_1), (x_2, y_2), \dots, (x_{N+1}, y_{N+1})$

$$a_1 + b_1x_1 + c_1x_1^2 = y_1$$

$$a_1 + b_1x_2 + c_1x_2^2 = y_2$$

$$a_2 + b_2x_2 + c_2x_2^2 = y_2$$

$$a_2 + b_2x_3 + c_2x_3^2 = y_3$$

$$\vdots$$

$$2c_1 = 0$$

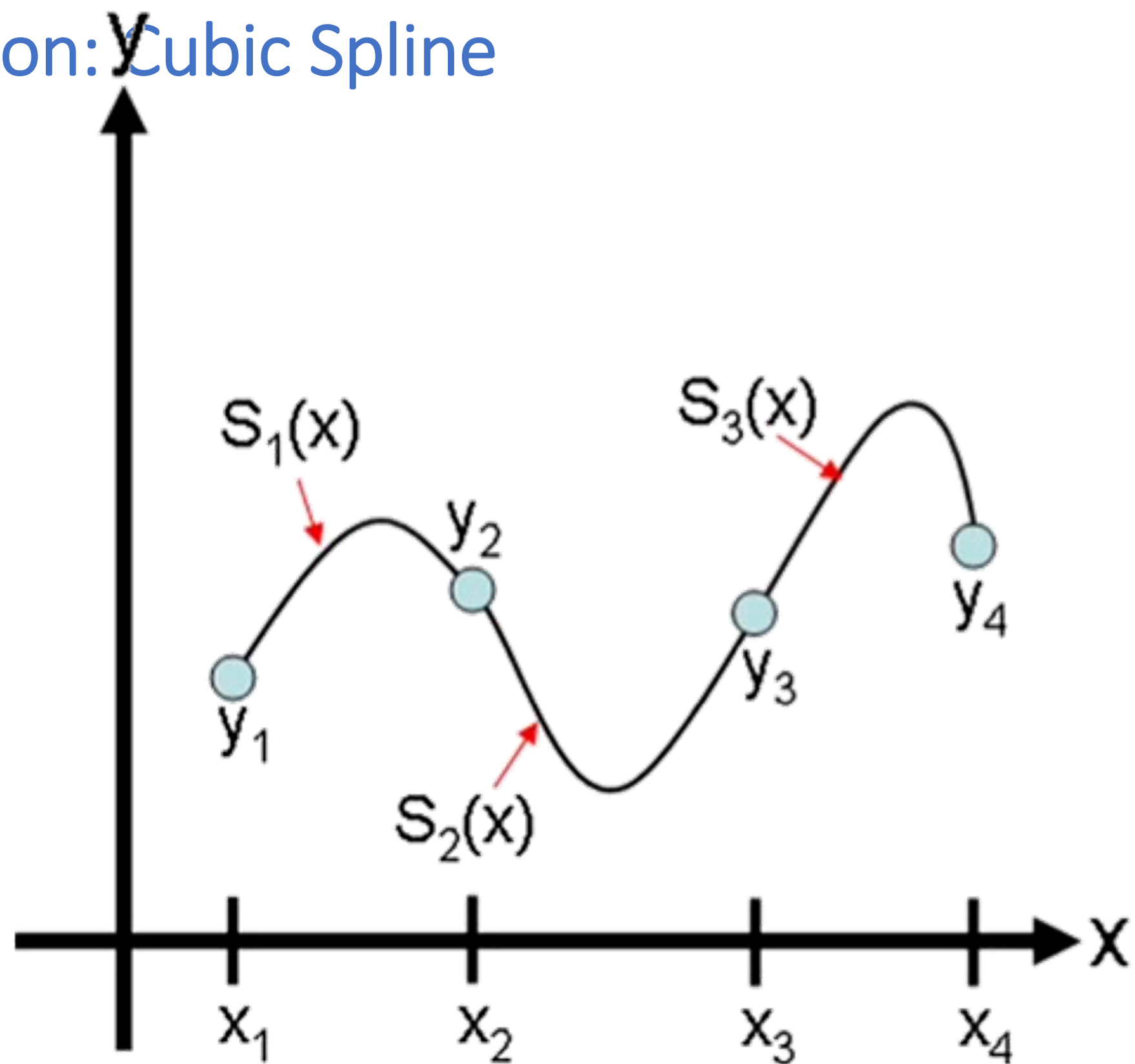
$$b_1 + 2c_1x_2 - b_2 - 2c_2x_2 = 0$$

$$b_2 + 2c_2x_3 - b_3 - 2c_3x_3 = 0$$

$$\vdots$$

Digital Image Interpolation: Cubic Spline

- Cubic spline interpolation

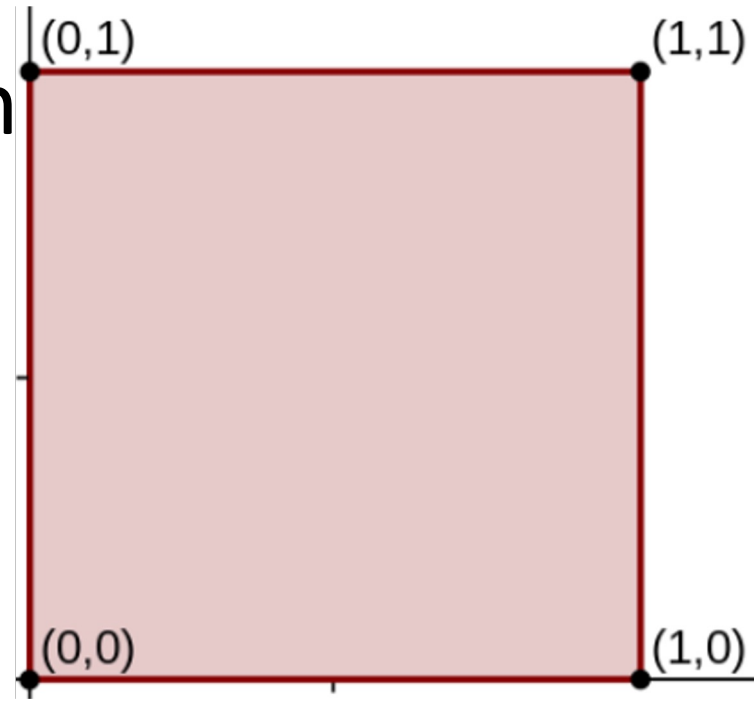


Digital Image Interpolation: Cubic Spline

- 1D case. **N intervals. Cubic** on each **interval**.
- Each cubic ($a_i + b_i x + c_i x^2 + d_i x^3$) has 4 parameters $\rightarrow$ total $4N$ unknowns
- **Constraints: within each interval**
 - Value of cubic must match data at left end & right end $\rightarrow N+N$ constraints
- **Constraints: at joints of intervals (total $N-1$ joints)**
 - First derivative of left and right cubic must match $\rightarrow N-1$ constraints
 - Second derivative of left and right cubic must match $\rightarrow N-1$ constraints
- **Constraints: boundary conditions**
 - e.g., design first and last polynomial to be quadratic ($d_1 = 0, d_N = 0$)
 - e.g., “natural” spline:
at left end of first interval specify 2nd derivative = 0 (curve $\rightarrow$ linear at end),
at right end of last interval specify 2nd derivative = 0 (curve $\rightarrow$ linear at end)

Digital Image Interpolation: Bicubic

- Given: Unit square with corner locations (0,0), (0,1), (1,0), (1,1). Values of function $f(x,y)$ & derivatives $f_x(x,y)$, $f_y(x,y)$, $f_{xy}(x,y)$ at 4 corners.
- Interpolated surface is a bicubic
 - 16 parameters
- At each **corner**, match **function values** (4 equations)
- At each **corner**, match function **derivatives** (3x4 equations)
- 16 linear equations. Solve.



$$p(x, y) = \sum_{i=0}^3 \sum_{j=0}^3 a_{ij} x^i y^j$$

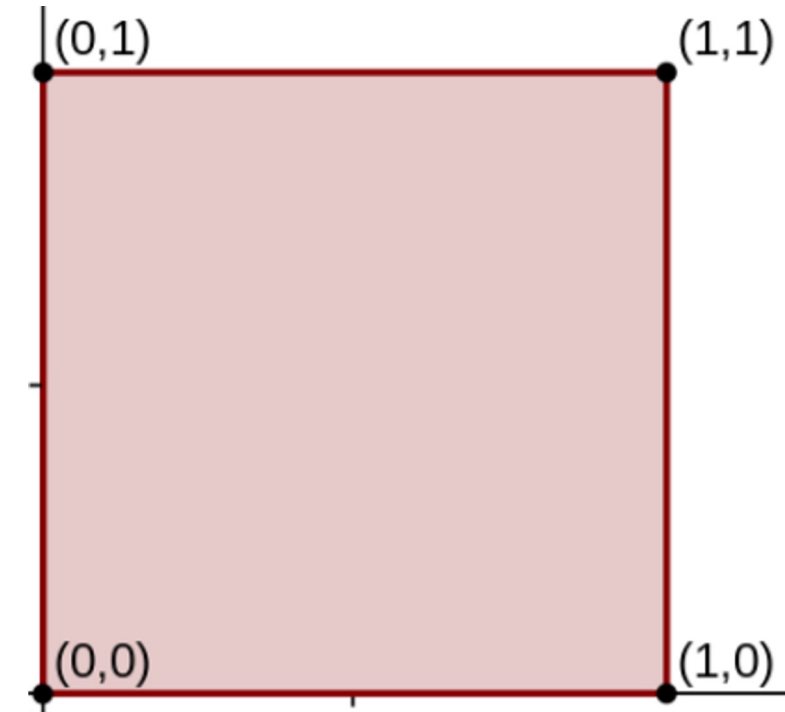
$$p_x(x, y) = \sum_{i=1}^3 \sum_{j=0}^3 a_{ij} i x^{i-1} y^j,$$

$$p_y(x, y) = \sum_{i=0}^3 \sum_{j=1}^3 a_{ij} x^i j y^{j-1},$$

$$p_{xy}(x, y) = \sum_{i=1}^3 \sum_{j=1}^3 a_{ij} i x^{i-1} j y^{j-1}$$

Digital Image Interpolation: Bicubic

- How to get function values and function-derivative values at corners ?
- Standard numerical schemes using finite-difference approximations
- e.g., $f_x(x,y) := 0.5 (f(x+1,y) - f(x-1,y))$
- e.g., $f_y(x,y) := 0.5 (f(x,y+1) - f(x,y-1))$
- e.g., $f_{xy}(x,y)$ can be obtained by finite-difference approximation of y derivative on $f_x(x,y)$, i.e.,
 $f_{xy}(x,y) := 0.25$
 $([f(x+1,y+1) + f(x-1,y-1)] - [f(x-1,y+1) + f(x+1,y-1)]) \dots$
- ... OR
finite-difference approximation of x derivative on $f_y(x,y)$



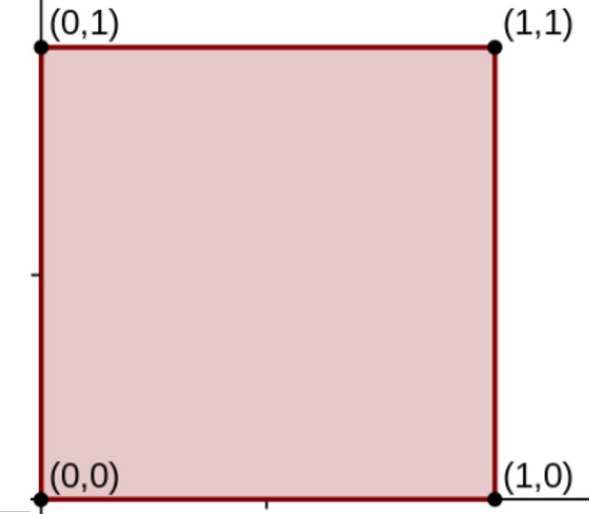
Digital Image Interpolation: Bicubic

- What about continuity of interpolated function across shared boundary ?
- Consider unit square. Consider interpolated function $p(x,y)$ over line segment joining (0,0) to (1,0)

1) $p(x, y=0)$,
 $p_x(x, y=0)$,
 $p_y(x, y=0)$,
 $p_{xy}(x, y=0)$

depend on which coefficients ?

- Coefficient used only when
 $j=0$ in $p(\cdot)$ & $p_x(\cdot)$ $j=1$ in $p_y(\cdot)$ & $p_{xy}(\cdot)$
- $\{a_{00}, a_{10}, a_{20}, a_{30}\} \cup \{a_{01}, a_{11}, a_{21}, a_{31}\}$



$$p(x, y) = \sum_{i=0}^3 \sum_{j=0}^3 a_{ij} x^i y^j$$

$$p_x(x, y) = \sum_{i=1}^3 \sum_{j=0}^3 a_{ij} i x^{i-1} y^j,$$

$$p_y(x, y) = \sum_{i=0}^3 \sum_{j=1}^3 a_{ij} x^i j y^{j-1},$$

$$p_{xy}(x, y) = \sum_{i=1}^3 \sum_{j=1}^3 a_{ij} i x^{i-1} j y^{j-1}$$

Digital Image Interpolation: Bicubic

- Consider interpolated function $p(x,y)$ over line segment joining $(0,0)$ to $(1,0)$
- 2) How do $\{a_{00}, a_{10}, a_{20}, a_{30}\} \cup \{a_{01}, a_{11}, a_{21}, a_{31}\}$ depend on data involving $(0,0)$ & $(1,0)$?
- Consider equations satisfied during bicubic fit involving $\{a_{00}, a_{10}, a_{20}, a_{30}\} \searrow$
- Consider equations satisfied during bicubic fit involving $\{a_{01}, a_{11}, a_{21}, a_{31}\} \searrow$
- Those 8 coefficients can be inferred (uniquely) from data at $(0,0)$ & $(1,0)$

$$p(x, y) = \sum_{i=0}^3 \sum_{j=0}^3 a_{ij} x^i y^j$$

$$p_x(x, y) = \sum_{i=1}^3 \sum_{j=0}^3 a_{ij} i x^{i-1} y^j,$$

$$p_y(x, y) = \sum_{i=0}^3 \sum_{j=1}^3 a_{ij} x^i j y^{j-1},$$

$$p_{xy}(x, y) = \sum_{i=1}^3 \sum_{j=1}^3 a_{ij} i x^{i-1} j y^{j-1}$$

$$f(0, 0) = p(0, 0) = a_{00},$$

$$f(1, 0) = p(1, 0) = a_{00} + a_{10} + a_{20} + a_{30}$$

$$f_x(0, 0) = p_x(0, 0) = a_{10},$$

$$f_x(1, 0) = p_x(1, 0) = a_{10} + 2a_{20} + 3a_{30}$$

$$f_y(0, 0) = p_y(0, 0) = a_{01},$$

$$f_y(1, 0) = p_y(1, 0) = a_{01} + a_{11} + a_{21} + a_{31}$$

$$f_{xy}(0, 0) = p_{xy}(0, 0) = a_{11},$$

$$f_{xy}(1, 0) = p_{xy}(1, 0) = a_{11} + 2a_{21} + 3a_{31}$$

Digital Image Interpolation: Bicubic

- Bicubic interpolation
- So far, we analyzed interpolated function over line segment joining $(0,0)$ to $(1,0) \rightarrow$ along X axis to find that $p(x,0)$, $p_x(x,0)$, $p_y(x,0)$, $p_{xy}(x,0)$ depend only on data at $(0,0)$ & $(1,0)$
- This will also hold true for other unit square with corners at $(0,0)$, $(1,0)$, $(1,-1)$, $(0,-1)$
- So, $p(x,y)$ and its partial derivatives with respect to x , y , xy are continuous across line segment
- But not necessarily second (non-mixed) partial derivatives, i.e., p_{xx} , p_{yy}
- Similar continuity properties will exist along all edges of square

Digital Image Interpolation: Tricubic

- Interpolating function
- 64 coefficients

$$f(x, y, z) = \sum_{i=0}^3 \sum_{j=0}^3 \sum_{k=0}^3 a_{ijk} x^i y^j z^k$$

- 1) Match function values at cube corners $\rightarrow$ 8 equations
- 2) Match function 1st derivatives (w.r.t. x, y, z) at cube corners
 $\rightarrow 8*3=24$ equations
- 3) Match function 2nd mixed derivatives (w.r.t. xy, yz, zx) at cube corners
 $\rightarrow 8*3=24$ equations
- 4) Match function 3rd mixed derivative (w.r.t. xyz) at cube corners
 $\rightarrow 8$ equations

INTERNATIONAL JOURNAL FOR NUMERICAL METHODS IN ENGINEERING
Int. J. Numer. Meth. Engng 2005; **63**:455–471

Published online 3 March 2005 in Wiley InterScience (www.interscience.wiley.com). DOI: 10.1002/nme.1296

Tricubic interpolation in three dimensions

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²Control and Dynamical Systems, California Institute of Technology, U.S.A.

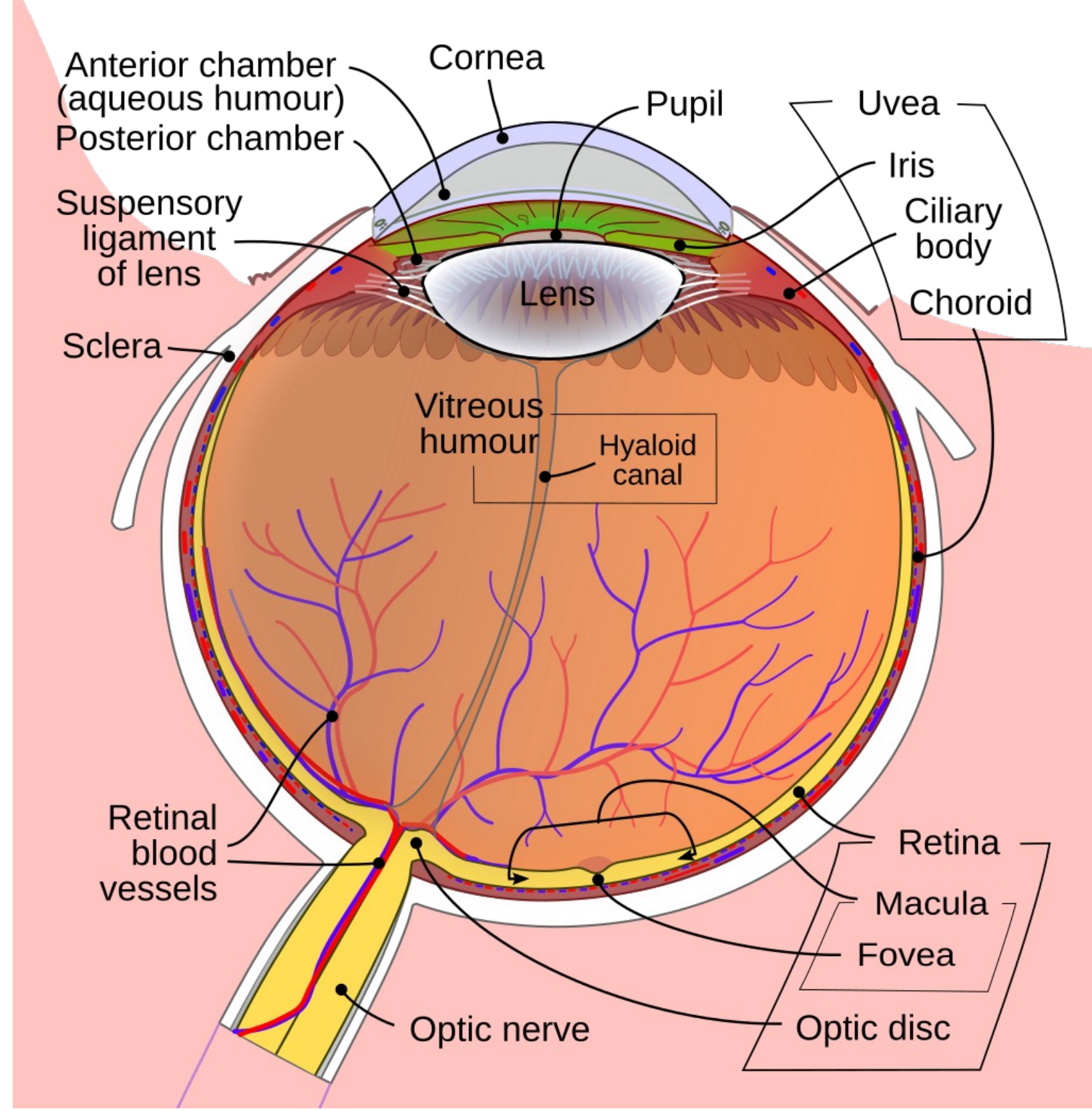
Digital Image Interpolation

- Image resizing
- Example 1: reduce image size
 - $301 \times 301 \rightarrow 101 \times 101$ may not need interpolation
 - $101 \times 101 \rightarrow 31 \times 31$ needs interpolation
 - Reducing image size
may introduce moire artifacts
- Example 2: increase image size along each dimension
 - Needs interpolation

Human Visual System

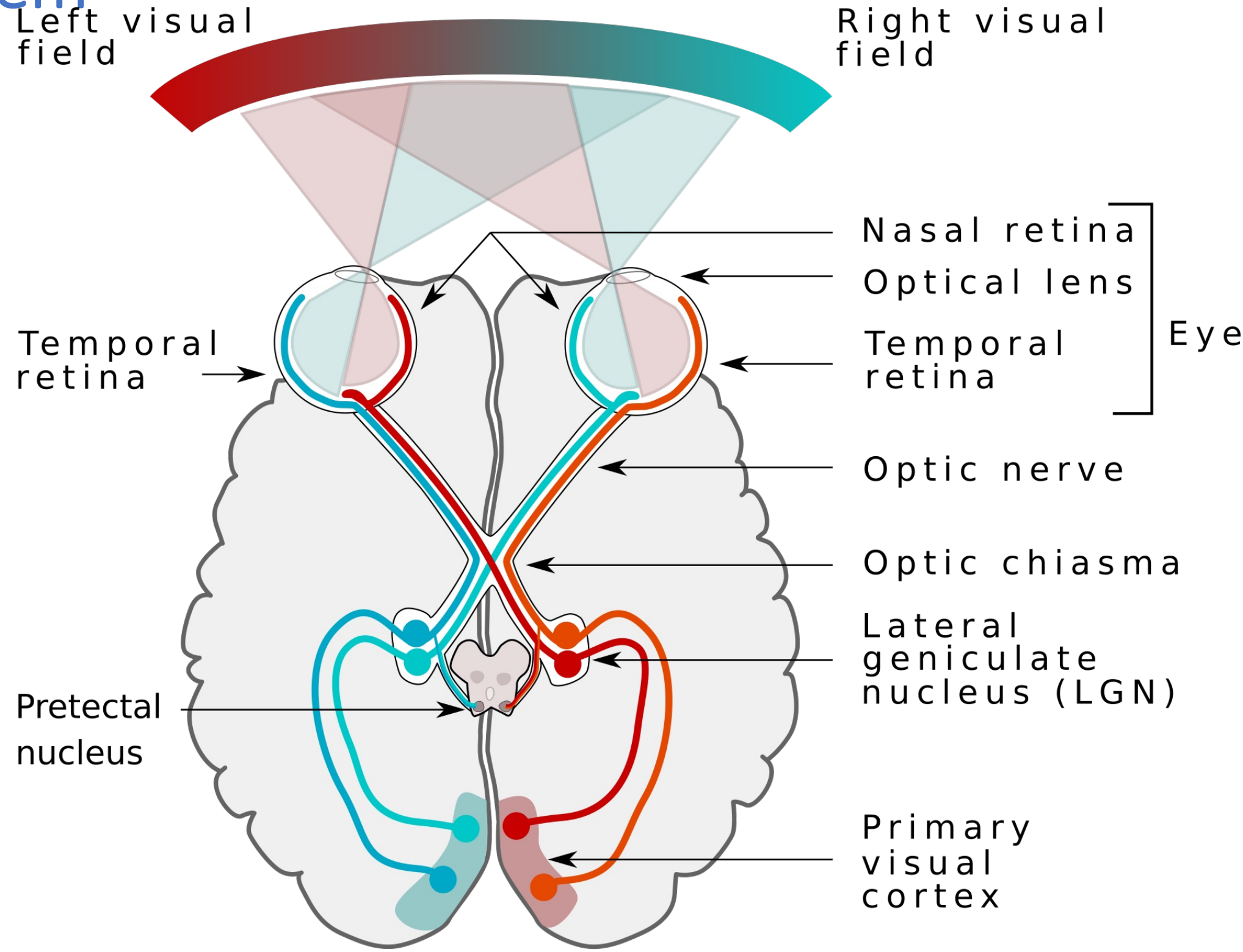
- Human eye

- en.wikipedia.org/wiki/Visual_system



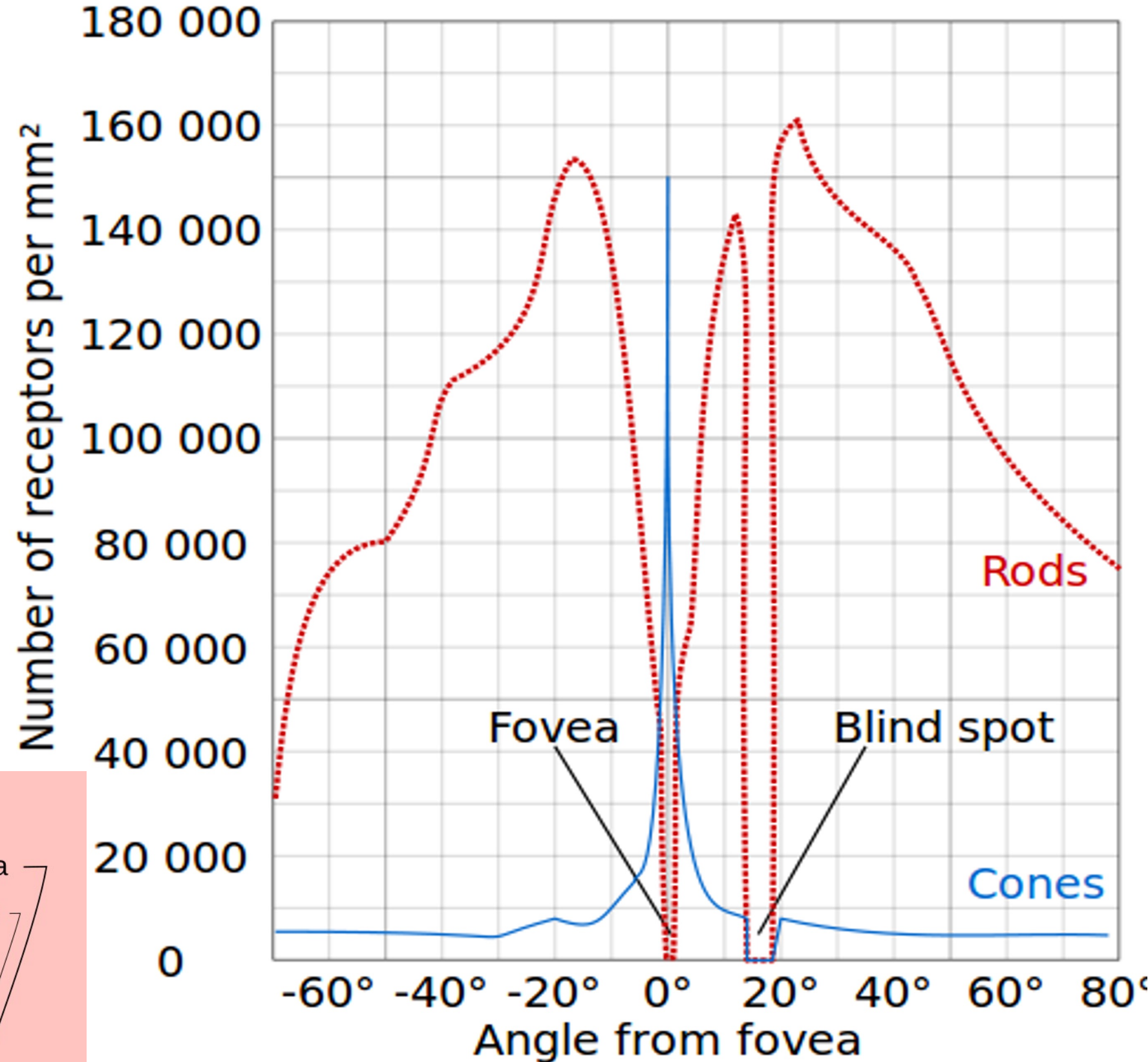
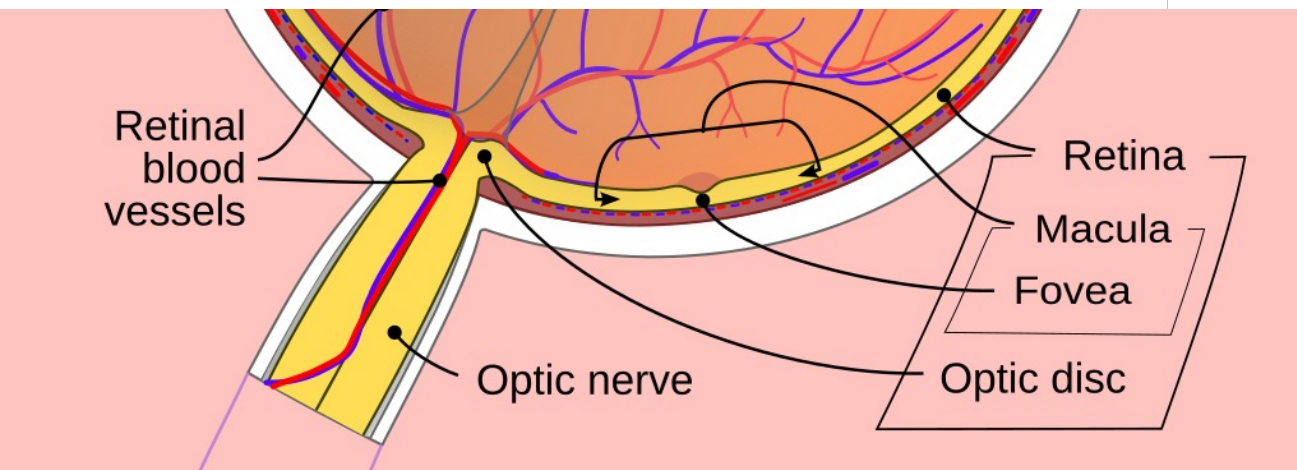
Human Visual System

- Visual interpretation in human brain



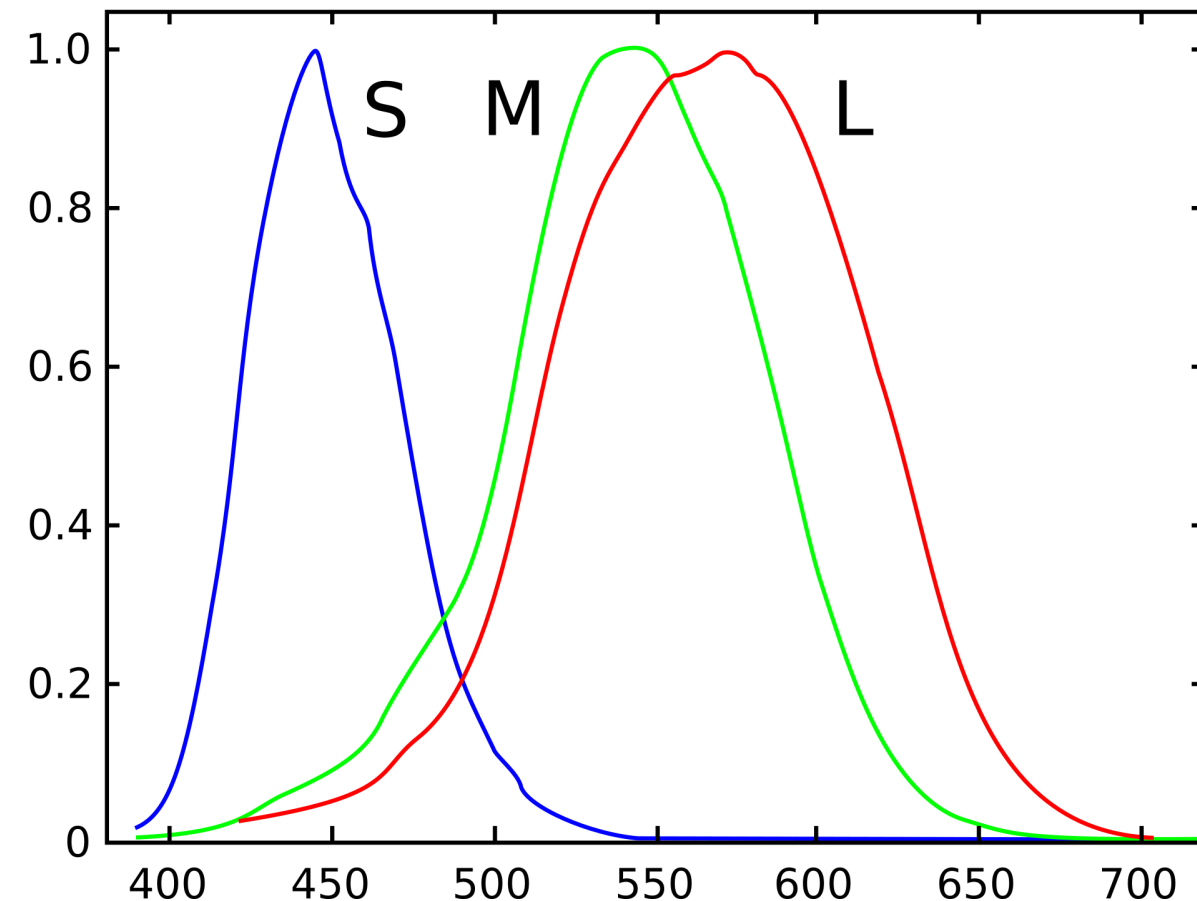
Human Visual System

- Human retinal cells
 - Cones → Color vision, visual acuity
 - Rods → Low-light vision, more sensitive than cones to intensity but NOT color



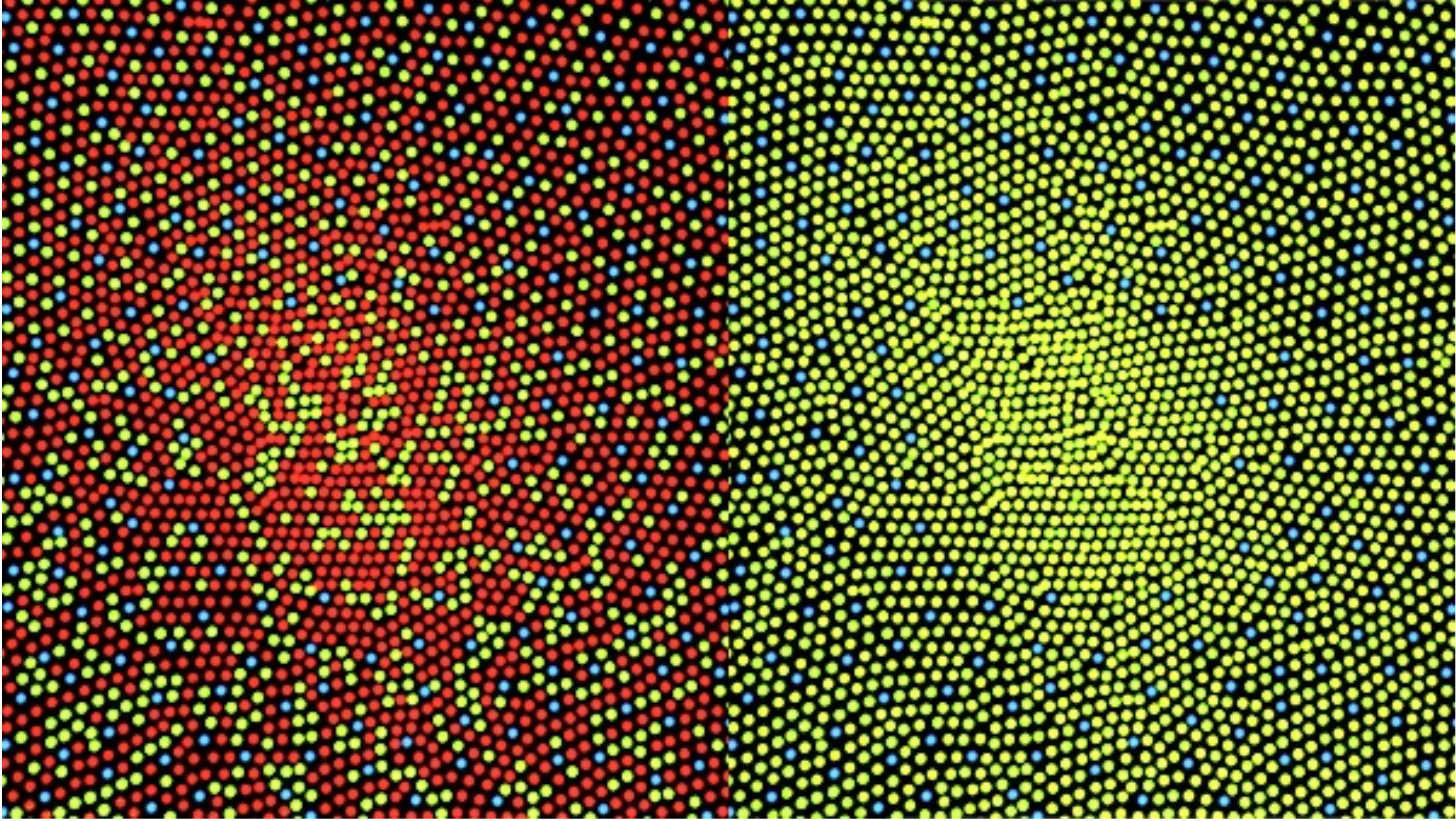
Human Visual System

- Human color vision is trichromatic
- Based on color-triangle theory proposed by Thomas Young; extended by James Clerk Maxwell, Hermann von Helmholtz
- 3 different types of cone cells in human eye → 3 independent channels for conveying color information
- 3 types of cones (L, M, S) have pigments that respond best to light of
long wavelength (around 560 nm),
medium wavelength (around 530 nm),
short wavelength (around 420 nm)
 - Normalized responsivity spectra of human cone cells →



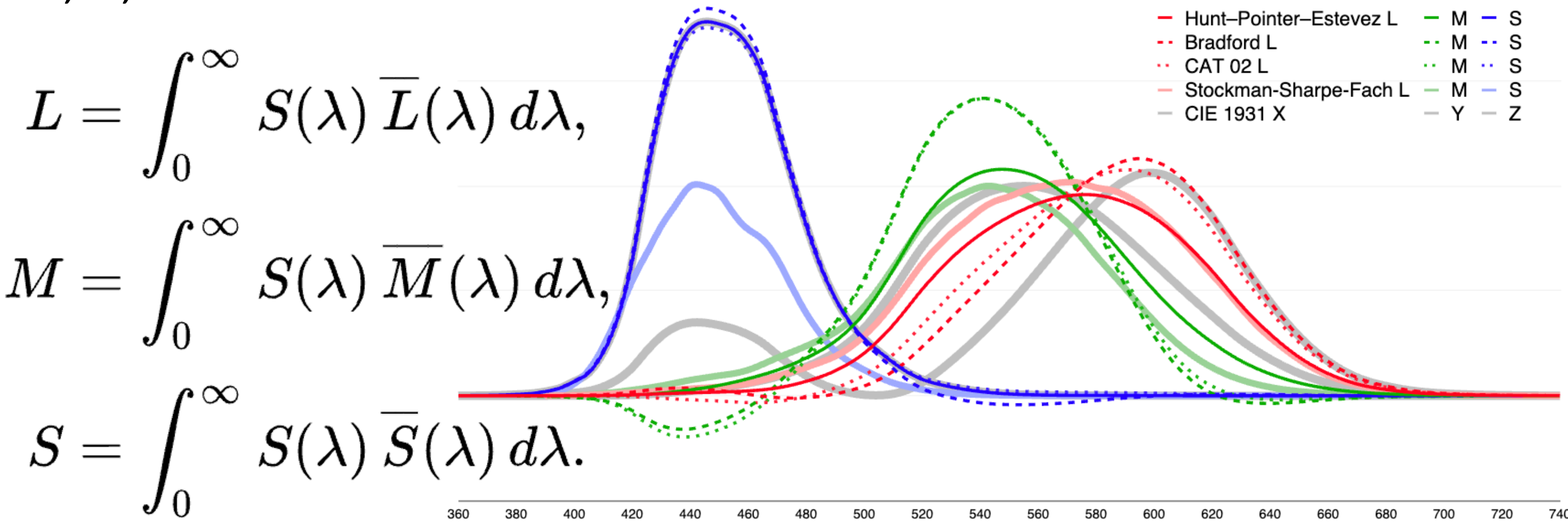
Human Visual System

- Human retina: distribution of cone cells in fovea
- Left:
normal
person
- Right:
color-blind
person



Human Visual System

- “LMS color spaces” (many definitions)
- Given incoming/stimulus light (into eye) of spectral density $S(\lambda)$
- Given spectral sensitivities of L,M,S cones as $\bar{L}(\lambda)$, $\bar{M}(\lambda)$, $\bar{S}(\lambda)$
- L,M,S cone-cell sensors measure:



Human Visual System

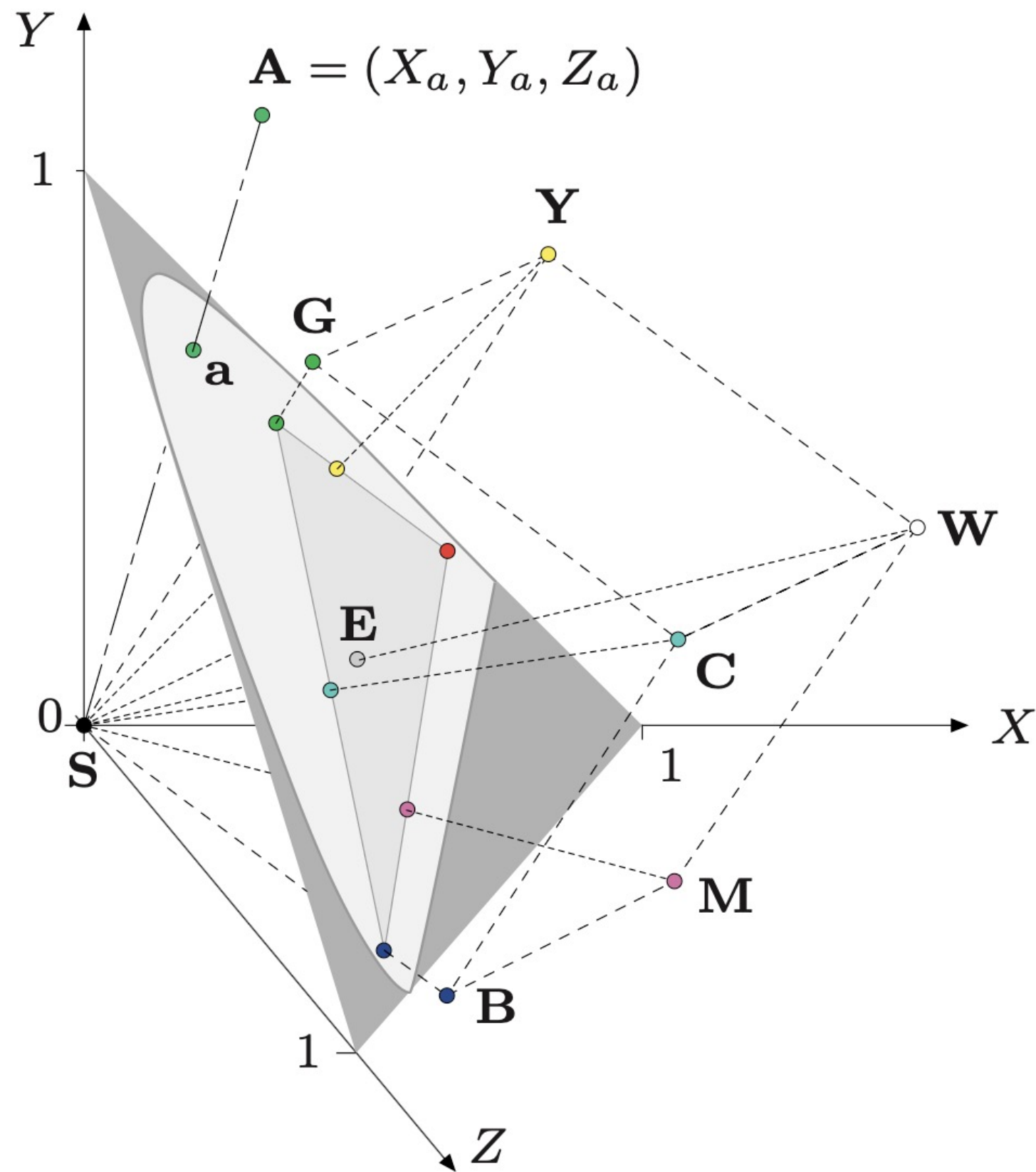


International Commission on Illumination
Commission Internationale de l'Eclairage
Internationale Beleuchtungskommission

- LMS color space relates to CIE XYZ color space
- Designs new spectral sensitivity functions $\bar{x}(\lambda)$, $\bar{y}(\lambda)$, $\bar{z}(\lambda)$...
- Also designs **new primary colors X, Y, Z** such that:
 - **All human-visible colors** (not representable in RGB space) can be modeled as a summation of **positive-only components**
→ all visible colors lie inside a 3D (generalized-)cone shaped region
 - Y component corresponds to perceived lightness or luminosity of a color
- Interestingly, visible colors don't include designed primary colors themselves → designed new primary colors X,Y,Z are **imaginary**
- Linear transformation between LMS and XYZ

Human Visual System

- New X,Y,Z imagined primary colors
- Defines 3 chromaticity values
 - $x := X / (X+Y+Z)$
 - $y := Y / (X+Y+Z)$
 - $z := Z / (X+Y+Z)$
 - White $(x,y,z) := (1/3,1/3,1/3)$
- RGB colors can be mapped to XYZ colors by linear transformation (not vice versa)



Human Visual System

- Visualize x, y . Ignore $z = 1 - x - y$.

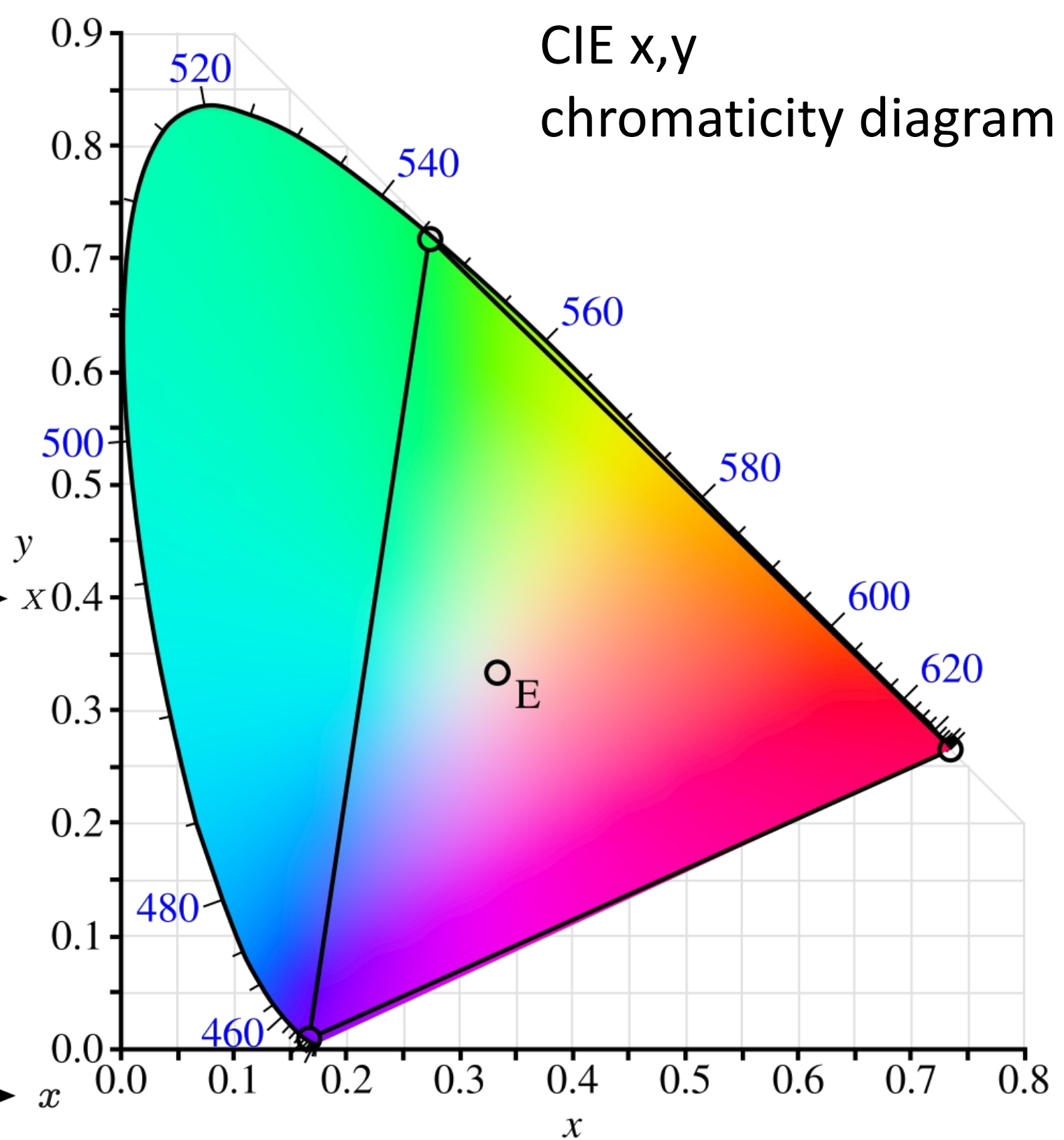
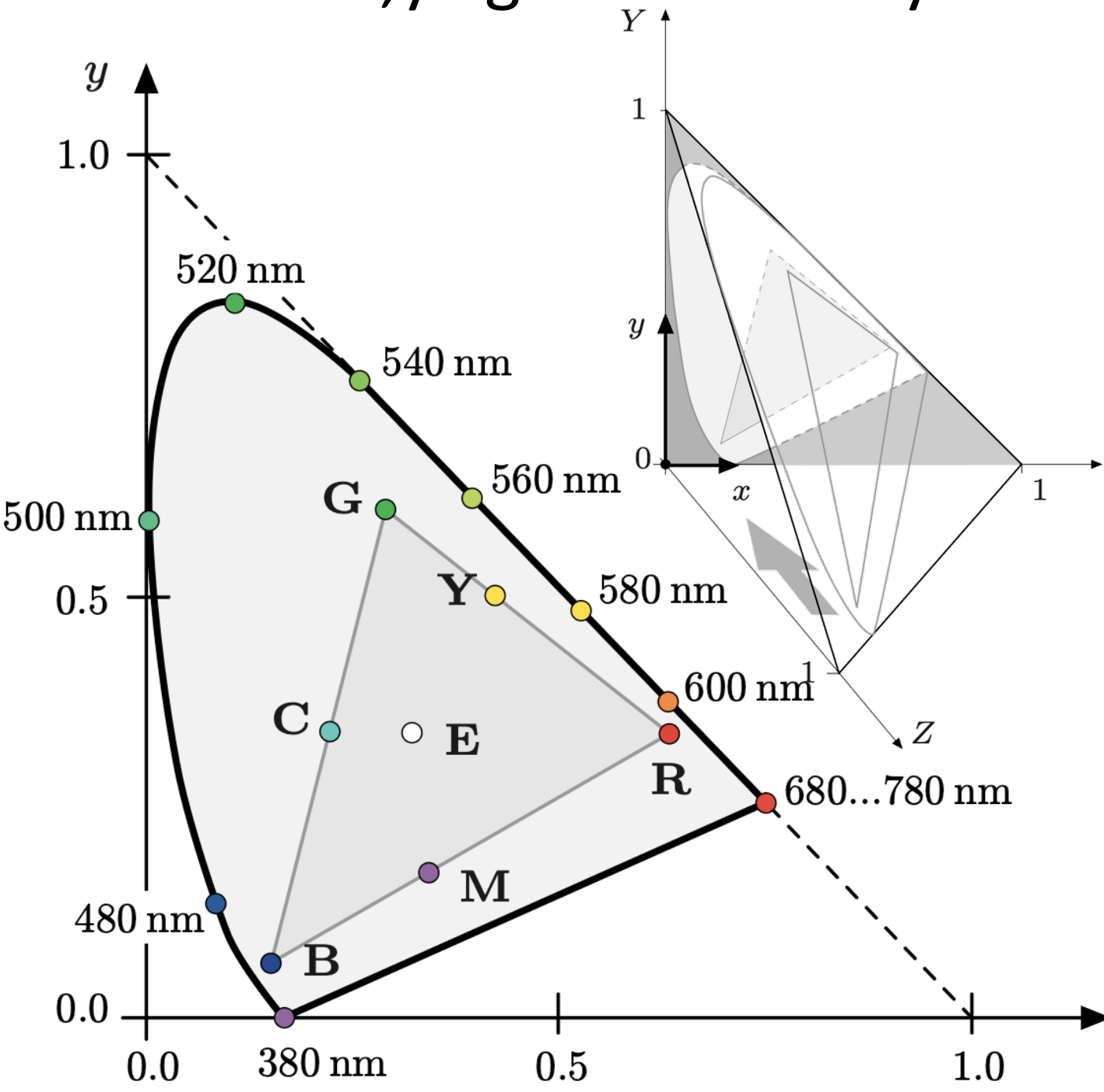
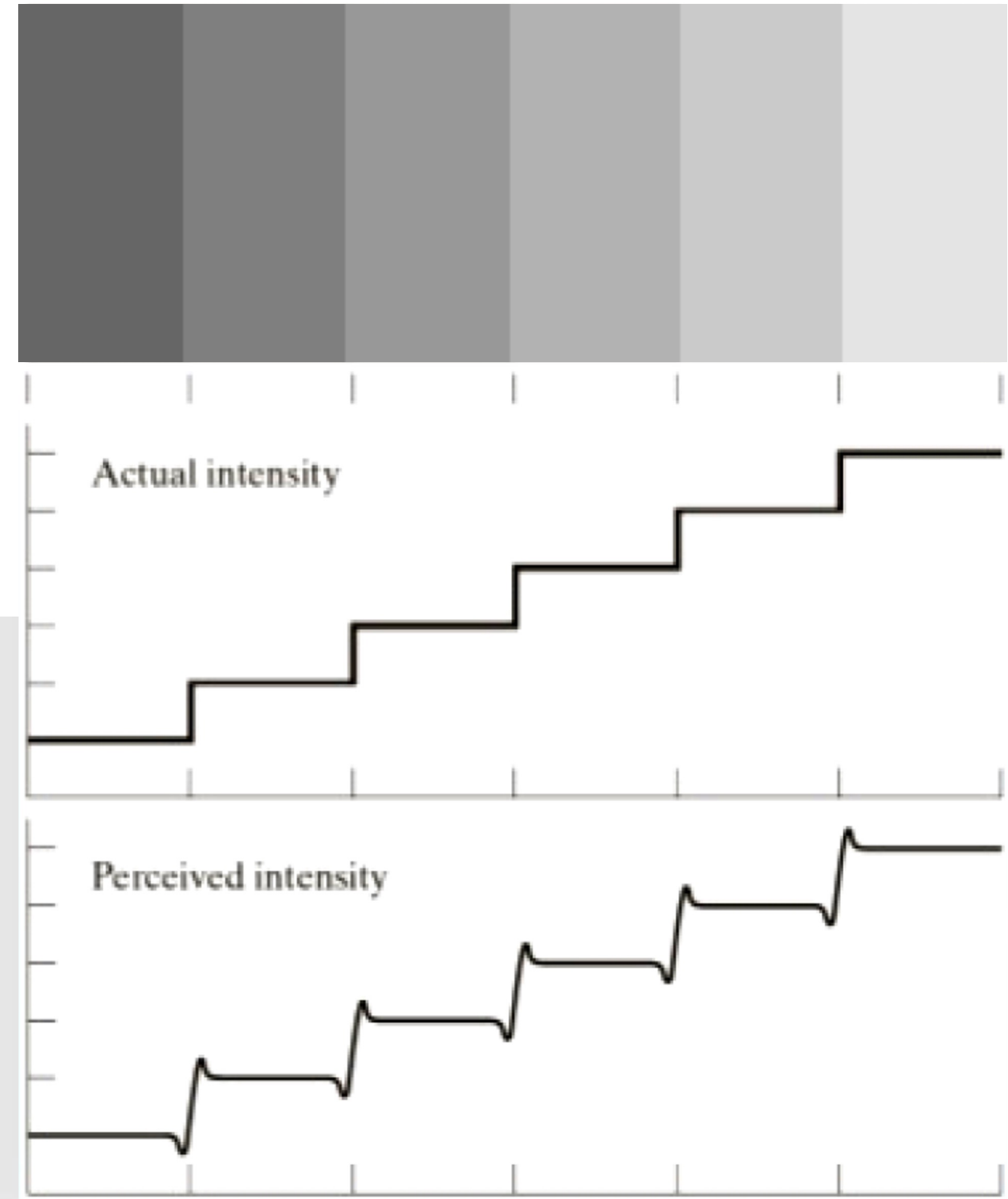


Image Perception: Optical Illusions

- “**Mach band**” effect
 - Perceived intensity NOT a simple function of actual intensity
 - Due to post-processing of visual signals by our visual system
- Exaggerated contrast between edges of slightly differing gray shades appears as soon as they touch



Ernst Mach

- Ernst Waldfried Josef Wenzel Mach
- Austrian physicist
 - Studied shock waves
→ Mach number
- Also worked in physiology, psychology (Mach bands)
- Philosopher: called “Buddha of science”

Mach band

Mach diamonds

Mach number

Mach reflection

Mach wave

Mach's principle

Criticism of Newton's
bucket argument^[1]

Empirio-criticism

Oblique effect

Relationalism

Shock waves

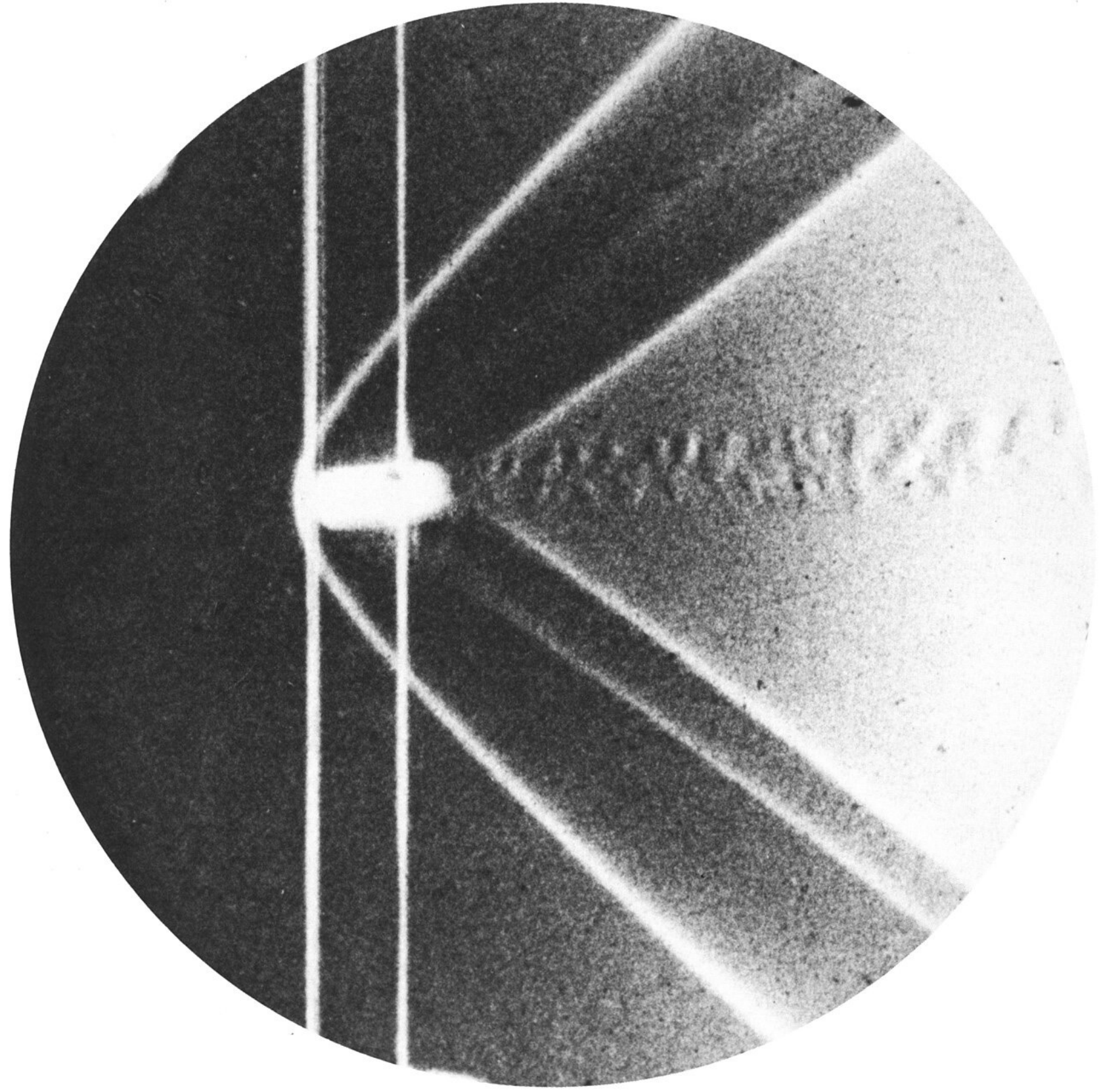
Stereokinetic stimulus

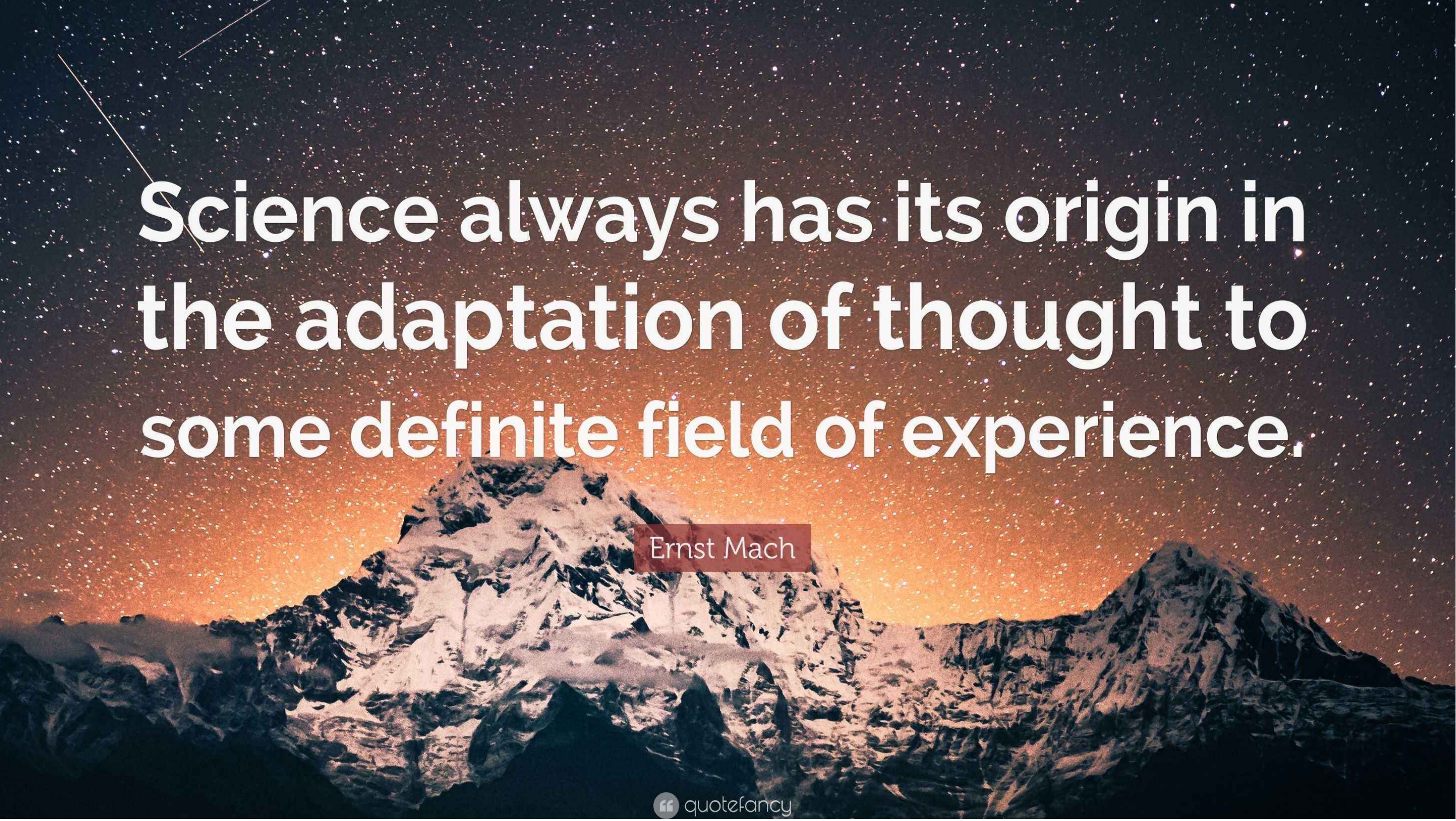
Ludwig Mach



Ernst Mach

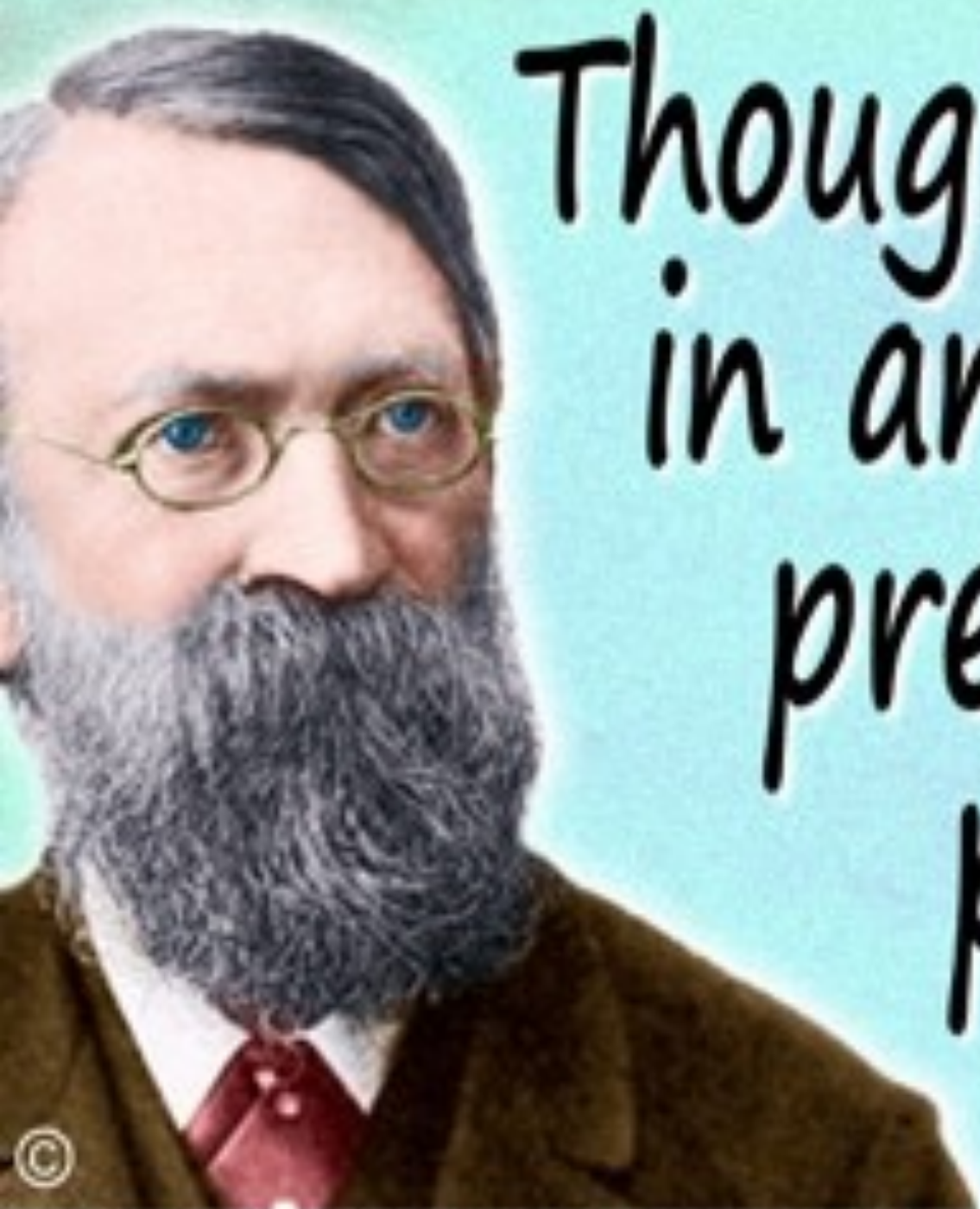
- Bullet, shock wave →
- Homeschooled until 14
- Godfather of
Wolfgang Ernst Pauli





Science always has its origin in
the adaptation of thought to
some definite field of experience.

Ernst Mach



Thought experiment is
in any case a necessary
precondition for
physical experiment.

Ernst Mach

Image Perception: Optical Illusions

- Mach band effect in chest X-rays

BMJ Case Reports

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Pseudo-pneumothorax: skin fold is an excellent imitator

[Kenichi Kishimoto](#)¹, [Takashi Watari](#)¹,  [Yasuharu Tokuda](#)²

Image Perception: Optical Illusions

- (A) Chest Xray shows curvilinear line at right lateral zone of right lung
- (B) Chest CT revealed absence of pneumothorax

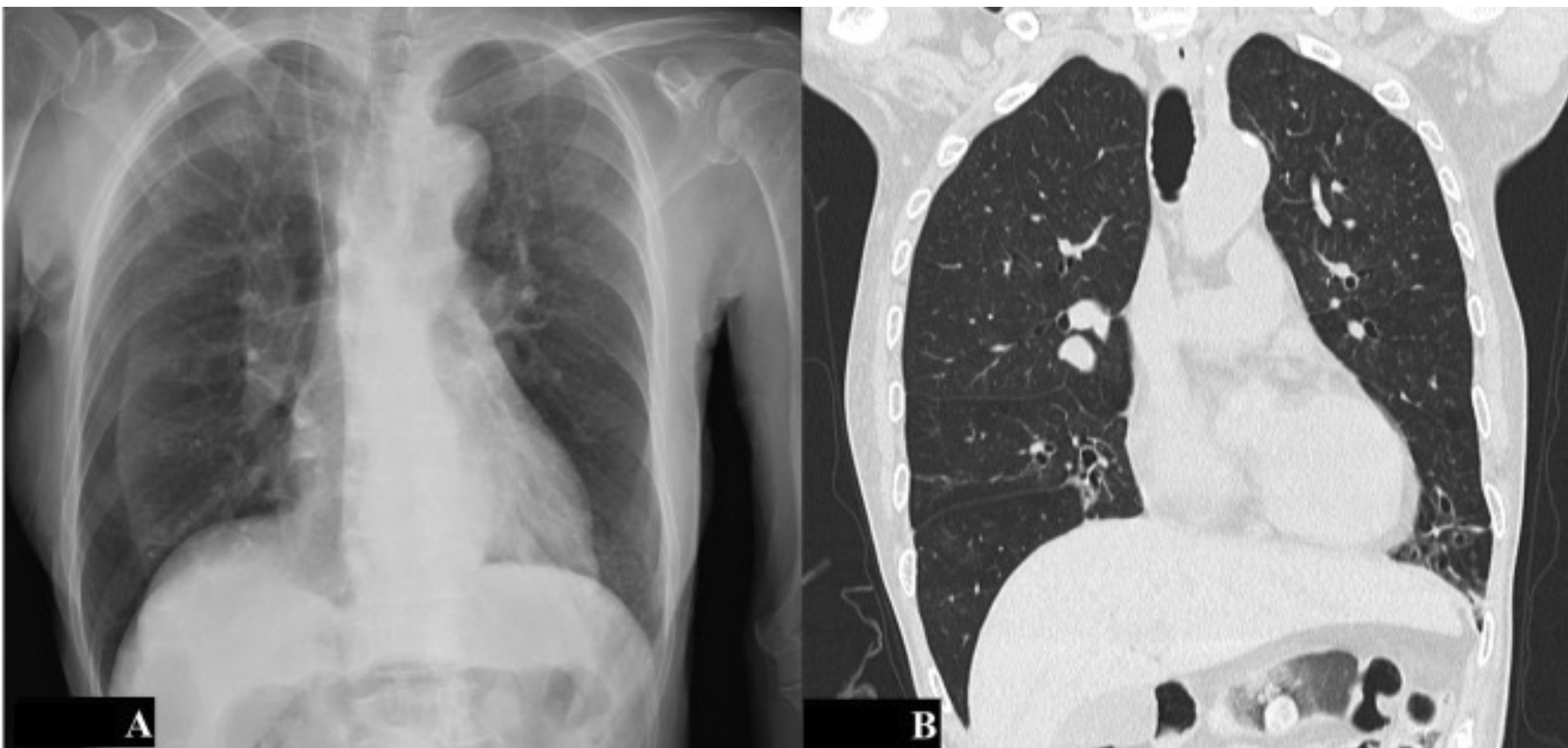


Image Perception: Optical Illusions

- “Dental students are more susceptible than dentists to misinterpreting as a horizontal root fracture (a Mach band illusion) what is known to be the junction of alveolar crestal bone and root.”
- “when presented with a scenario of trauma, both students and dentists are more likely to mistake what is being seen as a fracture line.”

JOURNAL OF ENDODONTICS
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Printed in U.S.A.
VOL. 27, No. 11, NOVEMBER 2001

Effect of Scenario and Experience on Interpretation of Mach Bands

Christen J. Nielsen, DMD, MS

Image Perception: Optical Illus

- Simultaneous contrast

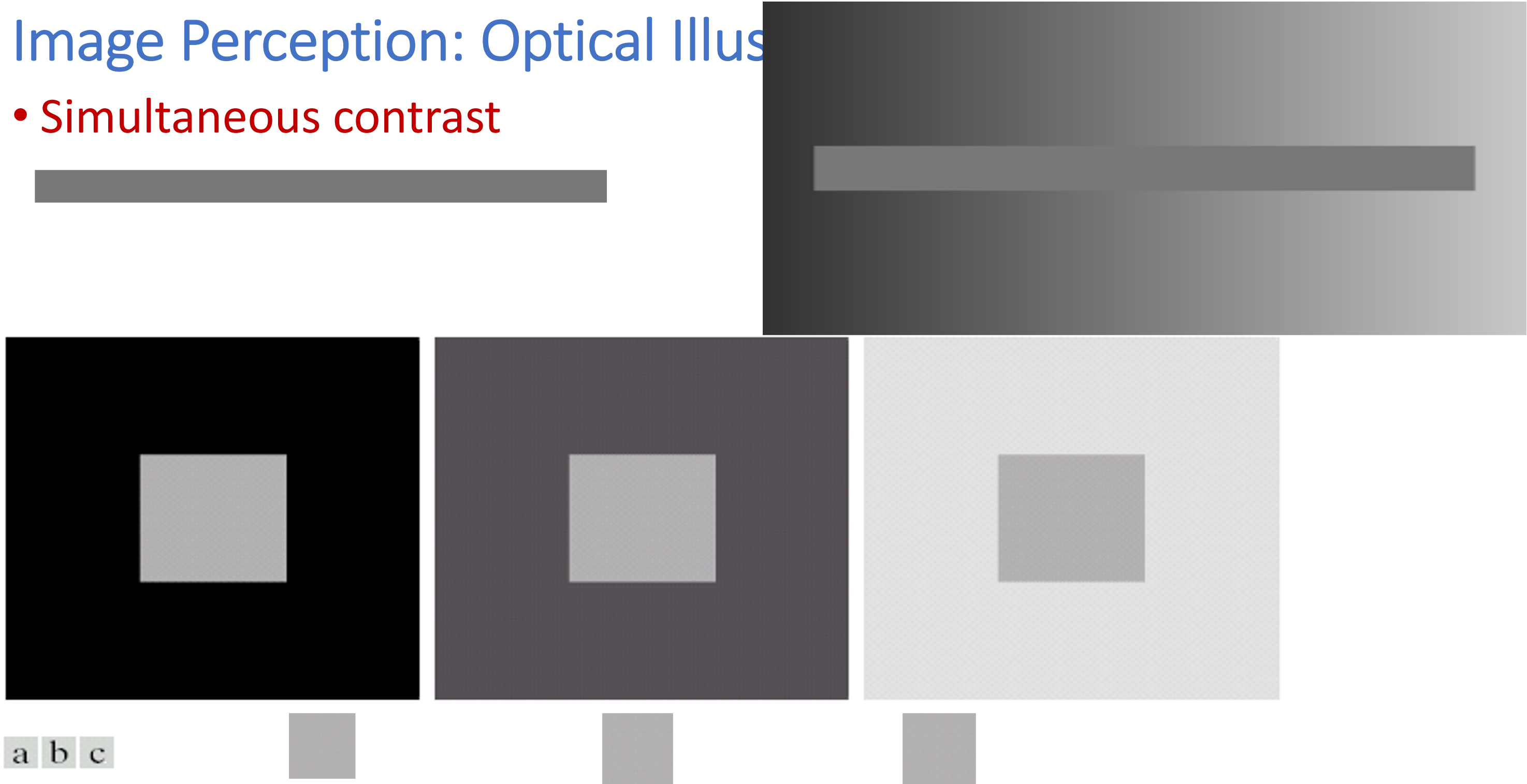


FIGURE 2.8 Examples of simultaneous contrast. All the inner squares have the same intensity, but they appear progressively darker as the background becomes lighter.

Image Perception: Optical Illusions

- Filling in missing data

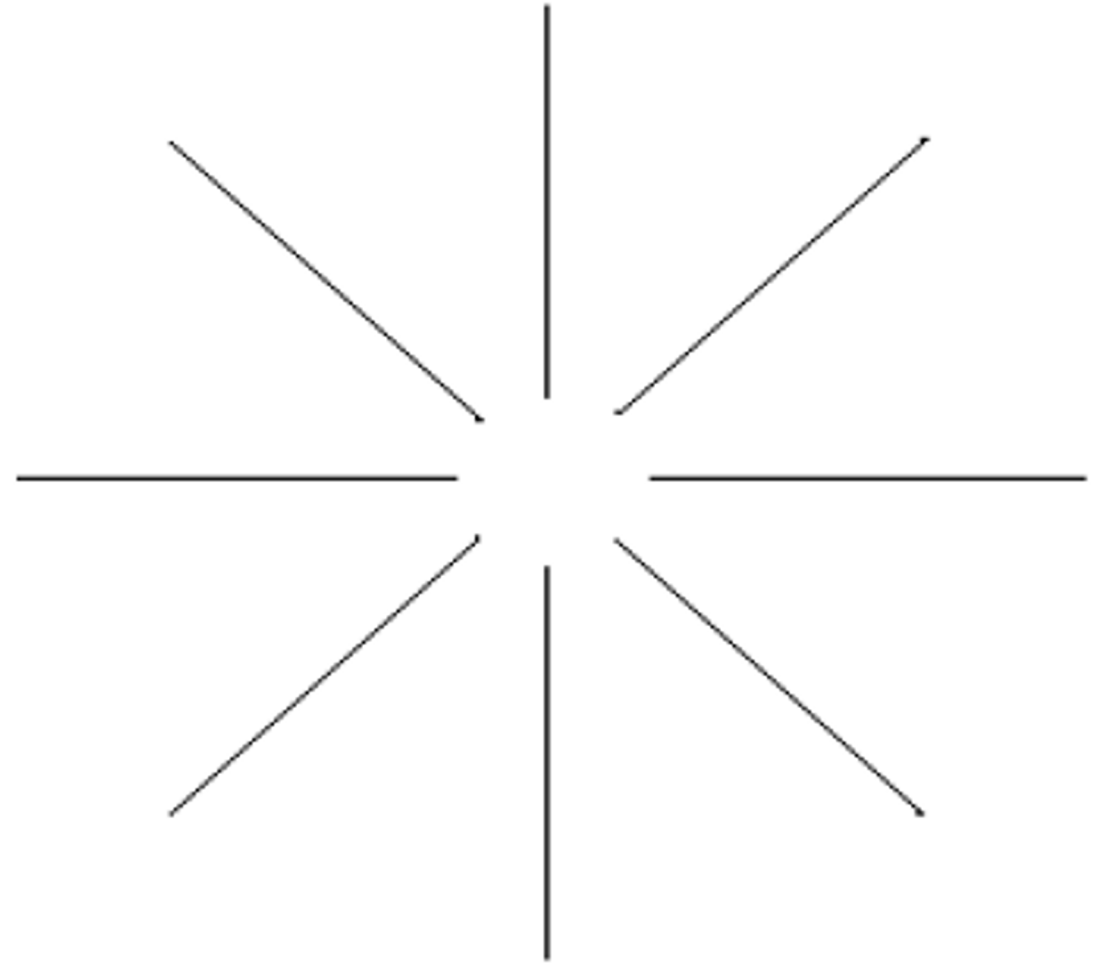


Image Perception: Optical Illusions

- Checkerboard illusion
- The squares marked A and B are ...
- ... the same shade of gray

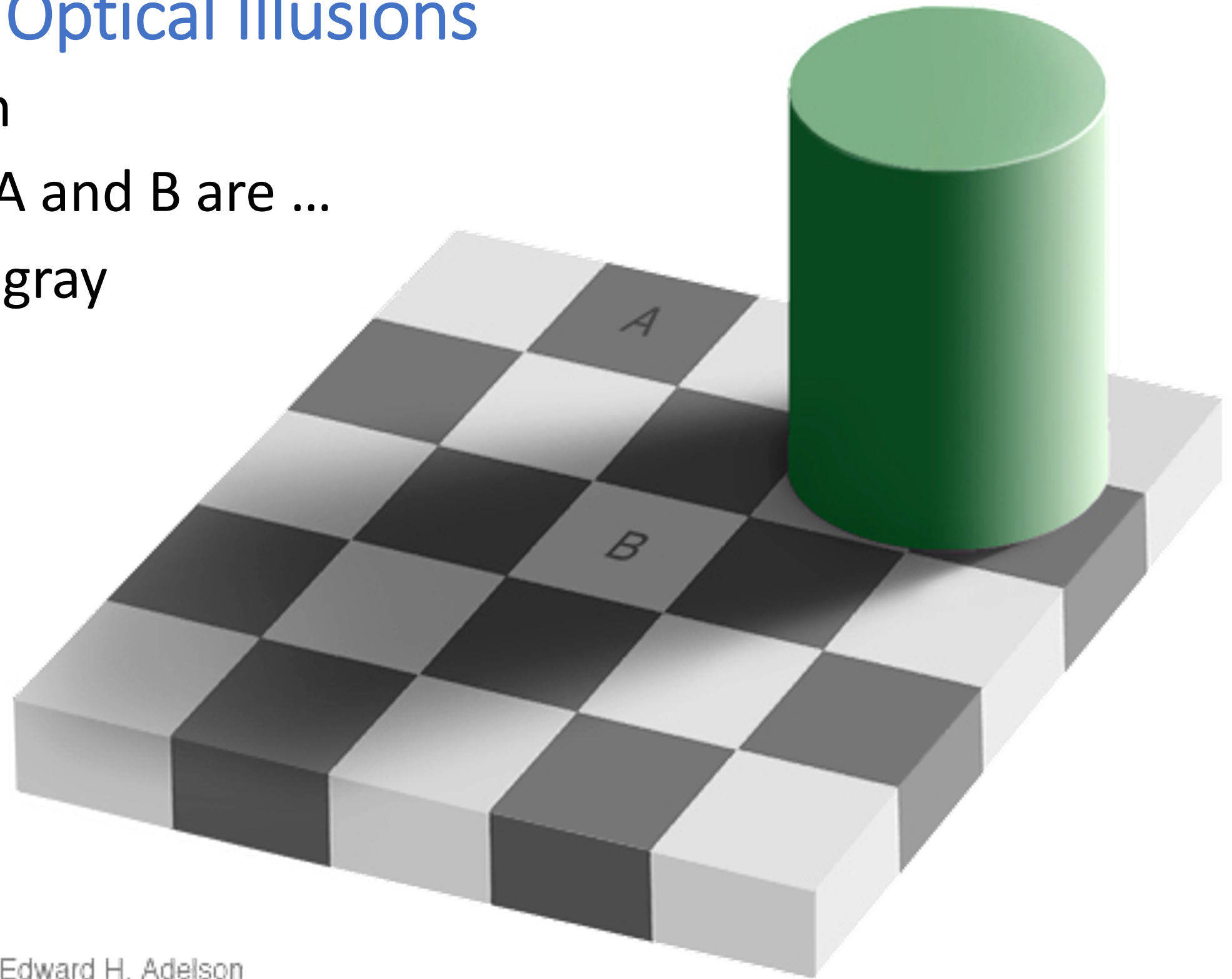


Image Perception: Optical Illusions

- Checkerboard illusion explanations:
 - 1) Simultaneous contrast
 - 2) Shadows
 - 3) Checkerboard pattern

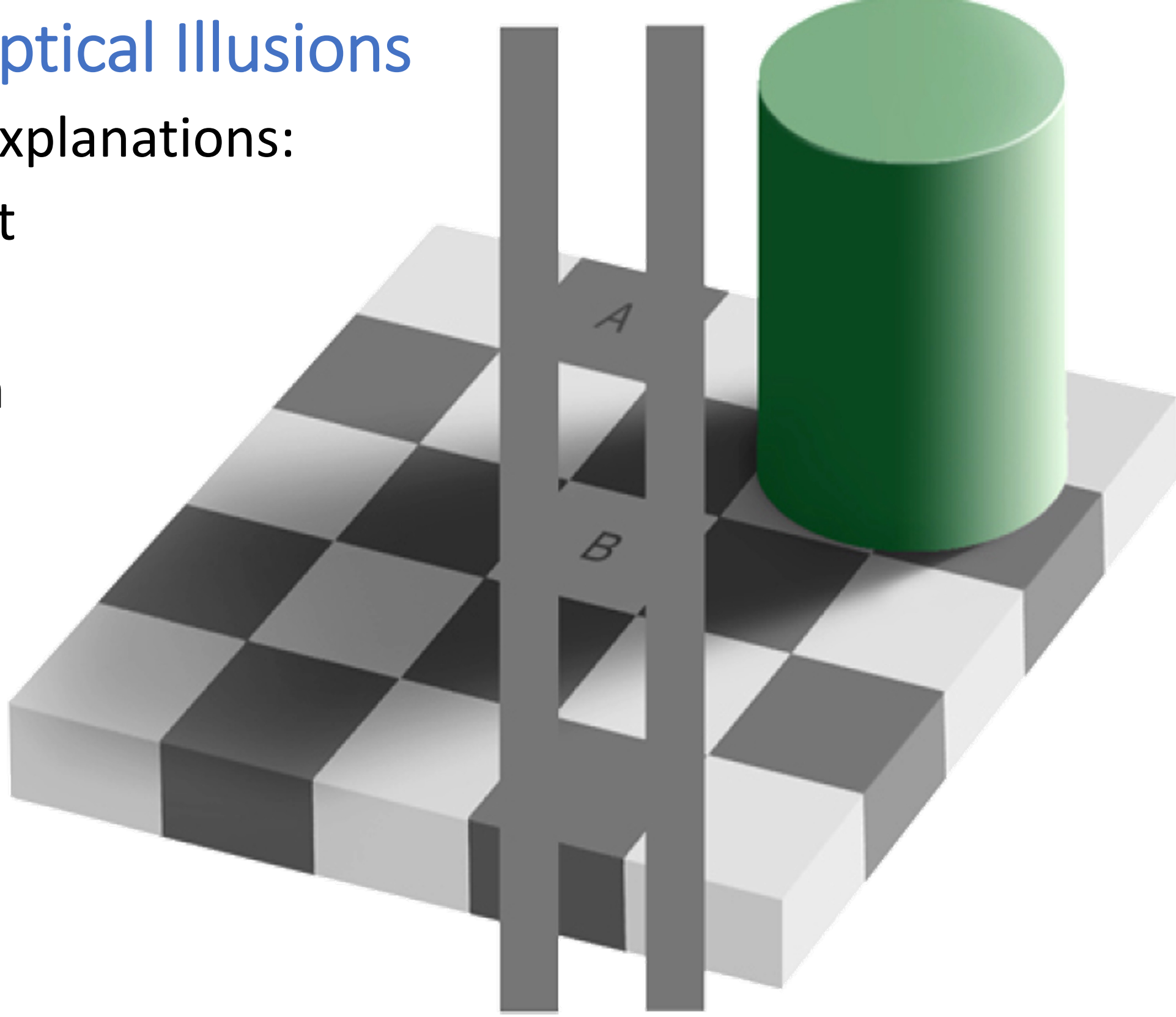


Image Perception: Optical Illusions

- Human visual system
- Is NOT optimized for measuring the amount of light
- Has evolved to achieve high-level understanding from limited data
- e.g., object detection and recognition despite occlusion
- e.g., object detection despite low light

- Be careful while interpreting results of your algorithms !